Term Paper

Relativistic Quantum Mechanics

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CERTIFICATE OF COMPLETION

This is to certify that ***Mr.* SAGAR JC**, student of St. Joseph's College (Autonomous), Bengaluru, bearing the register number 19PCM21015, has written the term paper titled ***Relativistic Quantum Mechanics*** in partial fulfillment of B.Sc. Course, under the guidance and supervision of ***Dr. Veena Adiga***, during the academic year 2021-2022.

Date: 26$^{\text{th}}$ November 2021

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DECLARATION

I Sagar J C, bearing the register number 19PCM21015 of St. Joseph's College (Autonomous), Bengaluru hereby solemnly declare that the term paper entitled "***Relativistic Quantum Mechanics***" submitted by me towards the partial fulfillment of my B.Sc. course to St. Joseph's College, Bengaluru is based on the information gathered from various sources primarily from standard books written on the same topic. All of which were reviewed, collected, studied & adopted into my term paper under the eminent guidance of Dr. Veena Adiga (Associate Professor), Dept. of Physics, St. Joseph's College (Autonomous), Bengaluru.

I also declare that the work embodied in the paper, either in part or in full, has not been previously submitted for any other Undergraduate Degree Course or in any other Journal or Publication.

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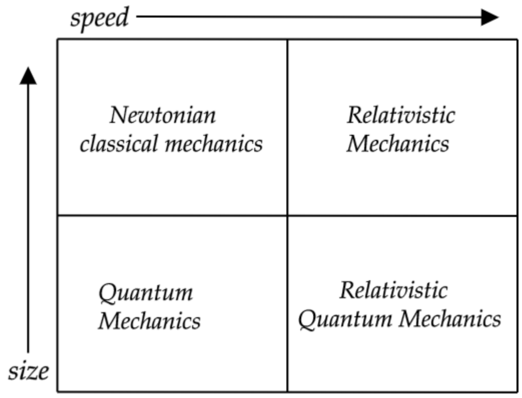
ABSTRACT

Relativity and Quantum Mechanics are the two main pillars of Modern physics. Both work perfectly fine in their domains. Quantum mechanics describes the behavior and physics of particles that have very small sizes and mass ($\leq 10^{-20}\ Kg$) perfectly. And Relativity comes into the picture when objects move at very high speeds ($\geq 0.1\ c$) comparable to the speed of light. But what about the particles that have very small sizes and move very fast? That is where the unification of Relativistic Mechanics and Quantum Mechanics becomes very important and gives rise to '***Relativistic Quantum Mechanics***', a new branch of modern physics where the rules of quantum mechanics abide by the principles of relativity.



In this term paper, I have reviewed the topic of Relativistic Quantum Mechanics. Starting from the prerequisites. Firstly special relativity, its notations, conventions, and some important results of relativistic mechanics. Then Quantum Mechanics, its postulates, the evolution of states, angular momentum, and spin. Then showing how the old quantum mechanics (Schrodinger's theory) had to be modified based on the condition of mass-energy equivalence. This gave rise to two main equations, Klein Gordon's equation and Dirac Equation. Then the paper focuses mainly on the Dirac theory of fermions, its consequences like how the spin of a particle emerged as a need to conserve angular momentum, the concept of anti-particles emerged to maintain the Lorentz invariance of Dirac Equation, the Dirac Lagrangian, various conserved quantities because of its symmetry, and the Dirac Sea theory (early attempts of quantum field theory) to explain the anti-matter particles, and finally on how matter and antimatter particles differ.

Contents

Chapter 1 : Introduction

1.1 Special Relativity

The *Theory of Special Relativity* was formulated by Albert Einstein in 1905 and made a profound impact on the understanding of Space and time. The two postulates put forth by Einstein in his paper were [1],

- Principle of Equivalence:- All the laws of physics (like Newton's laws, Maxwell's equations, etc) are the same for all the observers in any Inertial frame of Reference relative to one another.
- Universality of the speed of light:- Speed of light in a vacuum is the same for all observers in any frame of reference, regardless of their motion or the motion of the light source.

These two principles which were the foundations for special relativity broke the Galilean transformations. The Galilean transformations are given by the set of equations that are,

$$x' = x - v_x t \qquad y' = y \qquad z' = z \qquad \text{and} \qquad t' = t \tag{1.1}$$

The above Galilean transformations could not account for the universality of the speed of light. Consider,

Case 1:- If a light beam is traveling in the positive x-axis with a velocity 'c' as measured by the stationary observer O, then according to the O' observer who is moving in the same direction, using Eq.(1.1) the speed of light will be given by $u'_x = c - v_x$. Thus for the moving observer, if he is moving in the same direction of light then the speed of light has been reduced to u_x.

Case 2:- If a light beam is traveling in the negative x-axis with a velocity '$-c$' as measured by the stationary observer O, then according to the O' observer who is moving in the direction opposite to that of light, using Eq.(1.1), for him, the speed of light will be $u'_x = -c - v_x = -(c + v_x)$. Thus for the moving observer, if he is moving in the opposite direction of light then the speed of light has been increased to u_x which is $c+v_x$.

But according to Einstein's postulates, the speed of light has to be the same for all observers. So then to account for the Universality of Speed of Light, a new set of transformation rules was formulated called the '*Lorentz Transformations*' that related the position and time of a moving particle with respect to the stationary observer. They are given by [2],

$$\boxed{\begin{aligned} x' &= \gamma\,(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\left(t - \frac{xv}{c^2}\right) \end{aligned}} \qquad \text{where} \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{1.2}$$

Note that in this paper, the universal constants like the speed of light in vacuum c and the reduced Plank's Constant $\hbar$ appear very often in the equations, so I will be using the Natural Units setting $c = \hbar = 1$, Thus we will be getting the dimensional relation between various physical quantities as,

$$\boxed{[\text{Energy}] = [\text{Mass}] = [\text{Momentum}] = \frac{1}{[\text{Length}]} = \frac{1}{[\text{Time}]}} \tag{1.3}$$

Thus the Lorentz transformation for x' and t' can be written as,

$$x' = \gamma\,(x - vt) \qquad t' = \gamma\,(t - xv) \qquad \text{where} \qquad \gamma = \frac{1}{\sqrt{1 - v^2}} \tag{1.4}$$

1.1.1 Relativistic Notations

Four-vector

Four-vectors are vectors that have four components, one for time dimension and three for spatial dimension. The two types of four-vectors are Contravariant and Covariant, whose components are denoted by the Greek index. The components of a contravariant four-vector are denoted by a^μ (superscript) whose components are, $a^\mu = (a^0, a^1, a^2, a^3)$ and represented by a column matrix. Here, a^0 is the time component, and (a^1, a^2, a^3) are the x, y, z (spatial) components respectively. Similarly, the components of a covariant four-vector are denoted by a_μ (subscript) whose components are, $a_\mu = (a_0, a_1, a_2, a_3)$ and is represented by a row matrix. The contravariant four-vector (a^μ) and covariant four-vector (a_μ) are the duals of each other.

Metric signature and Metric tensor

The metric signature represents the signs of the non-zero diagonal elements of the metric tensor which helps in changing a covariant tensor into a contravariant tensor and vice versa.

$$\eta_{\mu\nu} \text{ is the Minkowski metric tensor where } \eta_{\mu\nu} = \eta^{-1}_{\mu\nu} = \eta^{\mu\nu} \text{ given by, } \eta_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{1.5}$$

The metric signature used here is $(1, -1, -1, -1)$. We can relate the components of a covariant and a contravariant four-vectors using the Metric Tensor. To change a contravariant tensor into a covariant tensor we have to multiply it by the metric tensor, $\eta_{\mu\nu}$. And to change a covariant tensor to a contravariant tensor we have to multiply it by the metric tensor, $\eta^{\mu\nu}$.

$$\therefore \qquad a_\mu = \eta_{\mu\nu}\, a^\nu \qquad \text{and} \qquad a^\nu = \eta^{\mu\nu}\, a_\mu \tag{1.6}$$

Thus, we get $(a^0 = a_0,\ a^1 = -a_1,\ a^2 = -a_2,\ a^3 = -a_3)$. So if $a^\mu = (a^0, a^1, a^2, a^3) = (a^0, \vec{a})$, then $a_\mu = (a_0, a_1, a_2, a_3) = (a^0, -a^1, -a^2, -a^3) = (a_0, -\vec{a})$.

Lorentz transformation of a Four-vector

Let a^μ be the four-vector according to the stationary observer in S frame and a'^μ be the four-vector according to the observer in S' frame who moving in $x-$axis at a constant velocity 'v'. Then the relation between the four-vectors a^μ and a'^μ is given by the Lorentz transformation rules given by,

$$a'^0 = \frac{a^0 - va^1}{\sqrt{1 - v^2}} \qquad a'^1 = \frac{a^1 - va^0}{\sqrt{1 - v^2}} \qquad a'^2 = a^2 \qquad a'^3 = a^3 \tag{1.7}$$

Inner product of four-vectors

Inner product is an operation that is always carried between a covariant and a contravariant four-vector. Let $a^\nu = (a^0, a^1, a^2, a^3)$, so the covariant form of the vector is given by, $a_\mu = \eta_{\mu\nu}\, a^\nu = (a^0, -a^1, -a^2, -a^3)$ and b^μ be a contravariant four-vector whose components are $b^\mu = (b^0, b^1, b^2, b^3)$. Then the inner-product between them is given by,

$$a_\mu b^\mu = a^0 b^0 - a^1 b^1 - a^2 b^2 - a^3 b^3 = a^0 b^0 - \vec{a} \cdot \vec{b} \tag{1.8}$$

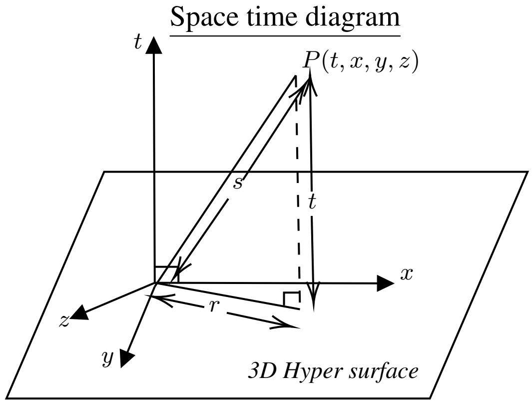
The magnitude of a four-vector $|a^\mu|$ is the distance of the four-vector in space-time. Its square is given by,

$$|a^\mu|^2 = a_\mu a^\mu = a^0 a^0 - a^1 a^1 - a^2 a^2 - a^3 a^3 = a^0 a^0 - \vec{a} \cdot \vec{a} \tag{1.9}$$

Note that any quantity which is of the form $a_\mu b^\mu$ is a scalar which means the corresponding components of the four-vectors are multiplied and then are summed where the spatial part will have a $-ve$ sign. So the inner-product yields a scalar which is a Lorentz invariant. So we get an identity,

$$a_\mu b^\mu = a^\mu b_\mu = a^0 b^0 - \vec{a} \cdot \vec{b} \tag{1.10}$$

1.1.2 Space-time coordinates and Space-time Interval



The coordinates of a point in space-time is given by the position four vector. The contravariant position vector is,

$$x^\mu = (x^0,\, x^1,\, x^2,\, x^3) = (t,\, x,\, y,\, z) \tag{1.11}$$

Its covariant form is, $$x_\mu = (x_0,\, x_1,\, x_2,\, x_3) = (x^0,\, -x^1,\, -x^2,\, -x^3) = (t,\, -x,\, -y,\, -z) \tag{1.12}$$

The space-time interval is got by the expression,

$$s^2 = x_\mu x^\mu = t^2 - x^2 - y^2 - z^2 = t^2 - r^2 \tag{1.13}$$

If the two events in space-time are very near then, the small four position vector can be denoted by $dx^\mu = (dx^0,\, dx^1,\, dx^2,\, dx^3) = (dt,\, dx,\, dy,\, dz)$ and the space-time interval between the two events is,

$$ds^2 = dx_\mu dx^\mu = dt^2 - dx^2 - dy^2 - dz^2 \tag{1.14}$$

Space-time intervals are categorized into three types based on the sign of 'ds^2'

1) Time like:- $ds^2 > 0 \quad \Rightarrow \quad dt^2 - dx^2 - dy^2 - dz^2 > 0 \quad \Rightarrow \quad dt^2 > dx^2 + dy^2 + dz^2$ (1.15a)
2) Space like:- $ds^2 < 0 \quad \Rightarrow \quad dt^2 - dx^2 - dy^2 - dz^2 < 0 \quad \Rightarrow \quad dt^2 < dx^2 + dy^2 + dz^2$ (1.15b)
3) Light like:- $ds^2 = 0 \quad \Rightarrow \quad dt^2 - dx^2 - dy^2 - dz^2 = 0 \quad \Rightarrow \quad dt^2 = dx^2 + dy^2 + dz^2$ (1.15c)

1.1.3 Relativistic Mechanics

Four Velocity

Four Velocity of a particle in space-time is defined as the rate of change of its fore-position with respect to its proper time and is denoted by $\vec{U}$ whose components are $U^\mu = (U^0, U^1, U^2, U^3) = (U^0, U^i)$ where U^0 is the temporal component of four-velocity and U^i are the spatial components of four velocity.

$$U^\mu = \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau}\right) \tag{1.16}$$

The rate of change of four position with proper time is equal to the speed of light.

$$\therefore \quad \frac{ds}{d\tau} = 1 \quad \Rightarrow \quad ds = d\tau = \sqrt{dt^2 - dx^2 - dy^2 - dz^2} \tag{1.17}$$

$$\Rightarrow \quad \frac{d\tau}{dt} = \sqrt{\frac{dt^2}{dt^2} - \frac{dx^2}{dt^2} - \frac{dy^2}{dt^2} - \frac{dz^2}{dt^2}} = \sqrt{1 - (v_x^2 + v_y^2 + v_z^2)} = \sqrt{1 - |\vec{v}|^2} = \frac{1}{\gamma} \quad \Rightarrow \quad \boxed{\frac{dt}{d\tau} = \gamma} \tag{1.18}$$

Note that the above equation predicts the phenomenon of time dilation which can be written as, $d\tau = \gamma dt$. Thus the 0^{th} component of four velocity is γ. Then the other three components are, $U^i = \left(\frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau}\right)$

$$U^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt} \cdot \frac{dt}{d\tau} = v^i \cdot \gamma = \gamma v^i \qquad \therefore \quad U^\mu = (\gamma, \gamma v_x, \gamma v_y, \gamma v_z) = (\gamma, \gamma\vec{v}) \tag{1.19}$$

Four Momentum

Four momentum or the relativistic momentum of a moving particle in space-time is given by $P_\mu = mU_\mu$ where m is the mass of particle and U_μ is the four velocity. The components of four momentum are,

$$P^\mu = \left(P^0,\ P^1\ P^2\ P^3\right) = \left(\frac{m}{\sqrt{1-v^2}}\ ,\ \frac{mv_x}{\sqrt{1-v^2}}\ ,\ \frac{mv_y}{\sqrt{1-v^2}}\ ,\ \frac{mv_z}{\sqrt{1-v^2}}\right) = m(\,\gamma, \gamma\vec{v}\,) = mU^\mu \tag{1.20}$$

The magnitude of the four momentum, $|\vec{P}|$ is a Lorentz invariant quantity given by the relation,

$$|\vec{P}|^2 = P_\mu P^\mu = m^2\gamma^2 \cdot (1-|v|^2) = m^2\gamma^2 \cdot \frac{1}{\gamma^2} = m^2$$

Thus the magnitude of the four momentum of a particle is, $\boxed{|\vec{P}| = m}$ or in SI units $\boxed{|\vec{P}| = mc}$ (1.21)

Relativistic Energy

Total energy of the particle is got by the Lagrange Mechanics and Lorentz transformations to be,

$$E = \frac{m}{\sqrt{1-v^2}} = P^0 \tag{1.22}$$

So the total energy of a particle is nothing but the zero-th component of four-momentum. So, we get,

$$P_\mu P^\mu = P_0^2 - P_x^2 - P_y^2 - P_z^2 = \frac{E^2 - p_x^2 - p_y^2 - p_z^2}{\sqrt{1-v^2}} = m^2 \tag{1.23}$$

For a particle moving slowly we have, $\quad v << 1 \quad \Rightarrow \quad E^2 - p^2 = m^2 \quad \Rightarrow \quad \boxed{E^2 = m^2 + p^2}$ (1.24)

This is the famous mass energy equivalence, in SI unit we get the equation to be, $\quad E^2 = m^2c^4 + p^2c^2$

Four Volume element

It is the product of the space-time coordinates which is similar to volume in 4 dimensions which is,

$$d\,\Omega = dx^0\,dx^1\,dx^2\,dx^3 = dt\,dx\,dy\,dz \tag{1.25}$$

Four Gradient

Four Gradient operator (∂) is the analogue of three gradient operator ($\vec{\nabla}$). There are two forms namely covariant four gradient (∂_μ) and contravariant four gradient (∂^μ).

Covariant $\quad\rightarrow\quad$ $$\partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_0,\ \partial_1,\ \partial_2,\ \partial_3) = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = \left(\frac{\partial}{\partial t},\ \vec{\nabla}\right) \tag{1.26}$$

Contravariant $\quad\rightarrow\quad$ $$\partial^\mu = \frac{\partial}{\partial x_\mu} = (\partial^0,\ \partial^1,\ \partial^2,\ \partial^3) = \left(\frac{\partial}{\partial t},\ -\frac{\partial}{\partial x},\ -\frac{\partial}{\partial y},\ -\frac{\partial}{\partial z}\right) = \left(\frac{\partial}{\partial t},\ -\vec{\nabla}\right) \tag{1.27}$$

$$\boxed{\partial_\mu = \left(\frac{\partial}{\partial t}, \vec{\nabla}\right)} \quad \text{and} \quad \boxed{\partial^\mu = \left(\frac{\partial}{\partial t}, -\vec{\nabla}\right)} \tag{1.28}$$

Four Laplacian operator - D' Alembert operator (D'Alembertian)

The four Laplacian (D'Alembertian), denoted by $\Box$ (which is the Box symbol), is the analogue of the Laplacian ∇^2 in 3D and is defined as,

$$\Box = \partial^\mu\partial_\mu = \sum_\mu \partial^\mu\partial_\mu = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} = \frac{\partial^2}{\partial t^2} - \nabla^2 \tag{1.29}$$

$$\Rightarrow \quad \boxed{\Box = \frac{\partial^2}{\partial t^2} - \nabla^2} \tag{1.30}$$

1.2 Non-Relativistic Quantum Mechanics (NRQM)

1.2.1 Mathematical Notations

Functions in Hilbert Space

In a finite dimension space, the vector $|\Psi\rangle$ has n-number of components. Consider a ket $|x\rangle$ which is an infinite dimensional vector whose components are all real numbers in the $x-$axis. Then a vector $|\Psi\rangle$ can be converted to a scalar function by the operation $\langle x|\Psi\rangle = \Psi(x)$. So, we can consider a function whose domain is the space and time coordinates which is denoted by $\Psi(t, x, y, z)$ to be an element of vector space.

- **The Hermitian conjugate of functions in Hilbert space**:- $\Psi^*(t, x, y, z)$ is the hermitian conjugate of $\Psi(t, x, y, z)$ which is got by substituting $i \to (-i)$ where ever it appears in the function.
- **Norm of a function is given by**:- $\langle\Psi|\Psi\rangle = \displaystyle\int_{-\infty}^{\infty} \Psi^*(t, x, y, z)\, \Psi(t, x, y, z)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z$ = Finite scalar.

  A function belongs to Hilbert space only if the norm is a finite scalar quantity which is square-integrable.
- **Inner product in Hilbert space**:- Inner product of two functions Φ and Ψ is given by,

$$\langle\Phi|\Psi\rangle = \int_{-\infty}^{\infty} \Phi^*(t, x, y, z)\, \Psi(t, x, y, z)\ \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z = \text{finite scalar} \tag{1.31}$$

- **Two functions are orthonormal if**:- $\langle\Psi_m|\Psi_n\rangle = \displaystyle\int_{-\infty}^{\infty} \Psi_m^*(t, x, y, z)\, \Psi_n(t, x, y, z)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z = \delta_{mn}$
- **Operators**:- In function space the operators are not matrices, rather they are mathematical tools like differentiation, etc. Examples:- $\dfrac{d}{dx}$, $\dfrac{\partial}{\partial t}$, $ia\dfrac{\partial^2}{\partial y^2}$, $\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + \dfrac{\partial^2}{\partial z^2}$, $\dfrac{\partial}{\partial t} + \dfrac{\partial}{\partial x} - 1$ etc.
- **Expectation value of an operator** = $\langle\Psi|\hat{\mathbf{A}}|\Psi\rangle = \displaystyle\int_{-\infty}^{\infty} \Psi^*(\hat{\mathbf{A}}\Psi)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z$ (Here Ψ is Normalized)

Eigen Value Equations

Consider an operator $\hat{\mathbf{A}}$ and a vector $|\Psi\rangle$, then if they satisfy the below condition,

$$\boxed{\hat{\mathbf{A}}|\Psi\rangle = \lambda|\Psi\rangle} \tag{1.32}$$

Where λ is a scalar quantity. The vector $|\Psi\rangle$ is called as the Eigen vector of $\hat{\mathbf{A}}$ and λ is the eigen value of the eigen vector Ψ of the operator $\hat{\mathbf{A}}$. And the whole equation is called as the Eigen value equation.

Some useful theorems

Theorem 1. *The Eigen values of a Hermitian operator are always real.*

Theorem 2. *The expectation value of a Hermitian operator corresponding to an eigen vector is real.*

Theorem 3. *Eigen functions of a Hermitian operator of distinct eigenvalues are mutually orthogonal.*

Theorem 4. *If two Hermitian operators commute with each other then they have the same eigen functions. And if two Hermitian operators don't commute then the eigen functions corresponding to them are different.*

Theorem 5. *Eigen vectors of a hermitian operator corresponding to distinct eigen values will form a complete set of mutually orthonormal basis vectors and the linear combination of these eigen vectors can represent any vector in Hilbert space.*

Theorem 6. *The eigen values and expectation values of an Anti-Hermitian operator are purely imaginary.*

Theorem 7. *Eigen values of a unitary operator are complex numbers whose magnitude is always* 1.

Theorem 8. *Eigen vectors of unitary operators are mutually orthogonal to each other provided that all the eigen values are distinct.*

1.2.2 Postulates of Quantum Mechanics

Postulate 1:- The state of the particle at a point in space (x, y, z) at some time t is represented by a state vector (or a wave function) denoted by $|\varPsi\rangle$ or $\varPsi(t, x, y, z)$ that belongs to Hilbert space. And the wave function is square-integrable, continuous (including its first and second derivatives) and tends to zero at boundary conditions or when $x, y, z \to (\infty, -\infty) \Rightarrow |\varPsi\rangle \to 0$

Postulate 2:- The probability of finding a particle of whose wave function is $|\varPsi\rangle$ in a region of space between x and $x + \mathrm{d}x$, y and $y + \mathrm{d}y$, z and $z + \mathrm{d}z$, i.e. within the volume element $(\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z)$ is given by the inner product of $|\varPsi\rangle$. This is Born's statistical interpretation of the wave function.

$$\text{Probability} = \langle\varPsi|\varPsi\rangle = \int_x^{x+\mathrm{d}x}\int_y^{y+\mathrm{d}y}\int_z^{z+\mathrm{d}z} \varPsi^*\varPsi\ \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \tag{1.33}$$

Thus the quantity $\varPsi^*\varPsi$ which we integrate upon space to find the probability is called Probability density. And since the particle if it exists it has to be somewhere in space, the Probability of finding it between $-\infty$ and ∞ should be 1. This is the Normalization condition given by, $\langle\varPsi|\varPsi\rangle = \int_{-\infty}^{\infty} \varPsi^*\varPsi\ \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = 1$

Postulate 3:- If a quantum operator is Hermitian then the operator is an observable quantity that can be measured. Hence to every observable quantity like position, momentum, energy, etc, there exists a corresponding Hermitian operator in quantum mechanics which has to satisfy the conditions,

$$\hat{\mathbf{A}}^\dagger = \hat{\mathbf{A}} \qquad \text{and} \qquad \langle\varPsi|\hat{\mathbf{A}}\varPsi\rangle = \langle\hat{\mathbf{A}}\varPsi|\varPsi\rangle \qquad \text{and} \qquad \langle\varPhi|\hat{\mathbf{A}}|\varPsi\rangle = \langle\varPsi|\hat{\mathbf{A}}^\dagger|\varPhi\rangle^* \tag{1.34}$$

Postulate 4:- The only possible values that can be obtained from the measurement of an observable of a system whose operator is $\hat{\mathbf{A}}$ are the eigen values a_n of the operator when operated on eigen functions $|\varPsi_n\rangle$.

$$\hat{\mathbf{A}}|\varPsi_n\rangle = a_n|\varPsi_n\rangle \qquad (\hat{\mathbf{A}} \text{ is the operator and } a_n \text{ is the observable quantity}) \tag{1.35}$$

Postulate 5:- The expectation value of an observable quantity is the average of the observed measurements. Hence, for a system that is represented by a normalized $|\varPsi\rangle$, the expectation value of the physical observable corresponding to $\hat{\mathbf{A}}$ (denoted by $\langle\hat{\mathbf{A}}\rangle$) is got by the inner product of $\varPsi$ with $\hat{\mathbf{A}}|\varPsi\rangle$.

$$\langle\hat{\mathbf{A}}\rangle = \langle\varPsi|\hat{\mathbf{A}}|\varPsi\rangle = \int_{-\infty}^{\infty} \varPsi^*(\hat{\mathbf{A}}\varPsi)\ \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \tag{1.36}$$

Postulate 6:- If $c_1|\varPsi_1\rangle$, $c_2|\varPsi_2\rangle$, $\ldots$, are wave functions of a system then $|\varPsi\rangle = c_1|\varPsi_1\rangle + c_2|\varPsi_2\rangle + \ldots$ is also a wave functions of the system subject to the condition $\sum_i c_i^2 = 1$ because of the normalization condition. This is called the principle of linear combination.

1.2.3 Time Evolution of a system and Hamiltonian as its Generator

Time evolution of a quantum system is the way how a system changes with time and what factors and quantities drive the change of the system. Let $|\varPsi\rangle$ and $|\varPhi\rangle$ be any two vectors that describe some quantum system. Then the inner product between them is given by $\langle\varPhi|\varPsi\rangle$ describes the projection of vectors on each other. We have to note that if two vectors are orthogonal to each other at a time t then even at some other time t' the vectors have to be orthogonal to each other. Similarly, if a vector is normalized, then it has to remain normalized with time. So the condition in the time evolution of a system is,

$$\langle\varPhi(t)|\varPsi(t)\rangle = \langle\varPhi(t')|\varPsi(t')\rangle \tag{1.37}$$

The transformation of a state vector has to be got by an operator. So let that operator be $\hat{\mathbf{U}}$. Thus,

$$|\Psi(t')\rangle = \hat{\mathbf{U}}|\Psi(t)\rangle \qquad \text{and} \qquad \langle\Phi(t')| = \langle\Phi(t)|\hat{\mathbf{U}}^\dagger \tag{1.38}$$

Taking the inner product of the transformed vectors, $\langle\Phi(t')|\Psi(t')\rangle = \langle\Phi(t)|\hat{\mathbf{U}}^\dagger\hat{\mathbf{U}}|\Psi(t)\rangle$ (1.39)

But by Eq.(1.37) we need the inner products to remain same with time. Thus,

$$\langle\Phi(t')|\Psi(t')\rangle = \langle\Phi(t)|\hat{\mathbf{U}}^\dagger\hat{\mathbf{U}}|\Psi(t)\rangle = \langle\Phi(t)|\Psi(t)\rangle \quad \Rightarrow \quad \hat{\mathbf{U}}^\dagger\hat{\mathbf{U}} = \hat{\mathbf{I}} \quad \Rightarrow \quad \hat{\mathbf{U}} \text{ is unitary.} \tag{1.40}$$

Thus the operator which evolves a state vector has to be a Unitary operator. Hence the unitary operators are those operators that deal with the transformation of the system that can preserve relationships. Let us denote the unitary operator as a function of time since here it deals with the time evolution. Then,

$$|\Psi(t+\delta t)\rangle = \hat{\mathbf{U}}(\delta t)|\Psi(t)\rangle \tag{1.41}$$

Now if there is no time evolution of the system, i.e., if $\delta t = 0$ then we need, $|\Psi(t+0)\rangle = \hat{\mathbf{U}}(0)|\Psi(t)\rangle = |\Psi(t)\rangle$ which implies that $\hat{\mathbf{U}}(0)$ does nothing to the system, hence, $\hat{\mathbf{U}}(0) = \hat{\mathbf{I}}$. Now let us find the expression for the Unitary operator $\hat{\mathbf{U}}(\delta t)$. Let δt be so small that its higher powers be neglected without much effect.

Then we can write $\hat{\mathbf{U}}(\delta t)$ as, $\hat{\mathbf{U}}(\delta t) = \hat{\mathbf{I}} + k\delta t\hat{\mathbf{H}}$ (1.42)

Where k is some constant, and $\hat{\mathbf{H}}$ is some operator. Using $\hat{\mathbf{U}}(\delta t)\hat{\mathbf{U}}^\dagger(\delta t) = I$ we get,

$$(\hat{\mathbf{I}} + k\delta t\hat{\mathbf{H}})(\hat{\mathbf{I}} + k\delta t\hat{\mathbf{H}})^\dagger = \hat{\mathbf{I}} \quad \Rightarrow \quad (\hat{\mathbf{I}} + k\delta t\hat{\mathbf{H}})(\hat{\mathbf{I}} + k^*\delta t\hat{\mathbf{H}}^\dagger) = \hat{\mathbf{I}} \tag{1.43a}$$

$$\Rightarrow \quad \cancel{\hat{\mathbf{I}}} + k\delta t\hat{\mathbf{H}} + k^*\delta t\hat{\mathbf{H}}^\dagger + \cancel{k^*k(\delta t)^2\hat{\mathbf{H}}^\dagger\hat{\mathbf{H}}} = \cancel{\hat{\mathbf{I}}} \quad \Rightarrow \quad k\hat{\mathbf{H}} + k^*\hat{\mathbf{H}}^\dagger = 0 \tag{1.43b}$$

Now to simplify the equation conventionally lets put $k = -i \;\; \Rightarrow \;\; k^* = i$ where $i = \sqrt{-1}$. This gives us an interesting result, i.e., $\hat{\mathbf{H}}^\dagger = \hat{\mathbf{H}}$ which implies it is a Hermitian operator that represents some physical observable quantity that drives the time evolution of the system. Then we get,

$$\hat{\mathbf{U}}(\delta t) = \hat{\mathbf{I}} - i\delta t\hat{\mathbf{H}} \tag{1.44}$$

Now let us find the time evolution equation of the system.

$$|\Psi(t+\delta t)\rangle = \hat{\mathbf{U}}(\delta t)|\Psi(t)\rangle = (\hat{\mathbf{I}} - i\delta t\hat{\mathbf{H}})|\Psi(t)\rangle = |\Psi(t)\rangle - i\delta t\hat{\mathbf{H}}|\Psi(t)\rangle \tag{1.45}$$

$$\Rightarrow \quad -i\hat{\mathbf{H}}|\Psi(t)\rangle = \frac{|\Psi(t+\delta t)\rangle - |\Psi(t)\rangle}{\delta t} \tag{1.46}$$

Taking the limit as $\lim_{\delta t\to 0}$ on RHS, $-i\hat{\mathbf{H}}|\Psi(t)\rangle = \lim_{\delta t\to 0}\frac{|\Psi(t+\delta t)\rangle - |\Psi(t)\rangle}{\delta t} = \frac{\partial|\Psi\rangle}{\partial t}$ (1.47)

Thus, we get the expression, $\hat{\mathbf{H}}|\Psi\rangle = i\frac{\partial|\Psi\rangle}{\partial t}$ (1.48)

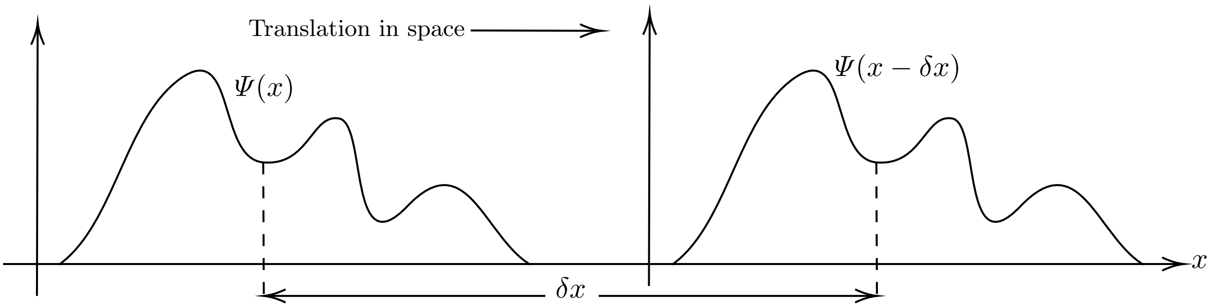
In classical mechanics, the time evolution of a system is driven by the Hamiltonian of the system. Thus the operator $\hat{\mathbf{H}}$ can be thought as the Hamiltonian operator. Then to match the equation dimensionally in SI units, we need a factor of $\hbar$ in the RHS. Thus we get,

$$\boxed{\hat{\mathbf{H}}|\Psi\rangle = i\hbar\frac{\partial|\Psi\rangle}{\partial t}} \tag{1.49}$$

This is the expression for the time evolution of a system which is driven by the Hamiltonian of the system and thus we get the Hamiltonian to be in the form of an operator as,

$$\hat{\mathbf{H}} = i\hbar\frac{\partial}{\partial t} \qquad \text{or in natural units setting } \hbar = 1 \text{ we get} \qquad \hat{\mathbf{H}} = i\frac{\partial}{\partial t} \tag{1.50}$$

1.2.4 Translation of a system and Momentum as its Generator



Let $\Psi(x)$ be the state vector of the system that depends on the position of the system. Then the translation of the system is got by the translational operator $\hat{\mathcal{T}}$ which has to be a unitary operator since only then the relationships between the state vectors of the system is preserved even after translation. So we have,

$$\hat{\mathcal{T}}(\delta x)\Psi(x) = \Psi(x-\delta x) \qquad (-ve \text{ sign since the wave function moves forward}) \tag{1.51}$$

Now if δx is 0, i.e., there is no translation, then Ψ remains the same. Hence we have $\hat{\mathcal{T}}(0) = \hat{\mathbf{I}}$. Now for very small δx, we can write the translation operator as, $\hat{\mathcal{T}}(\delta x) = \hat{\mathbf{I}} + k\delta x\hat{\mathbf{P}}$ where k is a constant and $\hat{\mathbf{P}}$ is some operator (The reason for it to be denoted as $\hat{\mathbf{P}}$ will be known after a few steps). Since the translational operator has to be Unitary, we get $\hat{\mathcal{T}}(\delta x)\hat{\mathcal{T}}^\dagger(\delta x) = \hat{\mathbf{I}}$. So,

$$(\hat{\mathbf{I}} + k\delta x\hat{\mathbf{P}})(\hat{\mathbf{I}} + k\delta x\hat{\mathbf{P}})^\dagger = \hat{\mathbf{I}} \qquad \Rightarrow \qquad k\hat{\mathbf{P}} + k^*\hat{\mathbf{P}}^\dagger = 0 \tag{1.52}$$

In the same manner, how we did in time evolution, let's conventionally put $k = -i$ and $k^* = i$ where $i = \sqrt{-1}$. This gives us $\hat{\mathbf{P}}^\dagger = \hat{\mathbf{P}}$ which implies it is a Hermitian operator and represents some observable which drives the translational motion of the system. Then we get $\hat{\mathcal{T}}(\delta x) = \hat{\mathbf{I}} - i\delta x\hat{\mathbf{P}}$. Now, Carrying out the similar steps as how we did before we get,

$$|\Psi(x-\delta x)\rangle = \hat{\mathcal{T}}(\delta x)|\Psi(x)\rangle = (\hat{\mathbf{I}} - i\delta x\hat{\mathbf{P}})|\Psi(x)\rangle = |\Psi(x)\rangle - i\delta x\hat{\mathbf{P}}|\Psi(x)\rangle \tag{1.53}$$

$$\Rightarrow \quad i\hat{\mathbf{P}}|\Psi(x)\rangle = \frac{|\Psi(x-\delta x)\rangle - |\Psi(x)\rangle}{-\delta x} \tag{1.54}$$

Taking the limit as $\lim_{\delta x\to 0}$ on RHS, $\quad i\hat{\mathbf{P}}|\Psi(x)\rangle = \lim_{\delta x\to 0}\dfrac{|\Psi(x-\delta x)\rangle - |\Psi(x)\rangle}{-\delta x} = \dfrac{\partial|\Psi\rangle}{\partial x} \qquad (1.55)$

Thus, we get the expression, $\quad \boxed{\hat{\mathbf{P}}|\Psi\rangle = -i\dfrac{\partial|\Psi\rangle}{\partial x}} \qquad (1.56)$

In classical mechanics, the translation of a system in space is driven by the Momentum of the system. Thus the operator $\hat{\mathbf{P}}$ can be thought as the Momentum operator which is the generator of Translation. Then to match the equation dimensionally in SI units, we need a factor of $\hbar$ in the RHS. Thus we get,

$$\boxed{\hat{\mathbf{P}}|\Psi\rangle = -i\hbar\frac{\partial|\Psi\rangle}{\partial x}} \qquad \text{where the momentum operator is,} \qquad \hat{\mathbf{P}} = -i\hbar\frac{\partial}{\partial x} \tag{1.57}$$

In natural units setting $\hbar = 1$ we get $\quad \hat{\mathbf{P}} = i\dfrac{\partial}{\partial x}\quad$. Classically we know that the Hamiltonian of the system is related to position and momentum by,

$$H = \frac{p^2}{2m} + V \tag{1.58}$$

Where p is the momentum, m is the mass of the system and V is the Potential energy of the system. Thus we get the quantum analog to them where the physical quantities are substituted by the respective operators. Thus we get,

$$i\hbar\frac{\partial}{\partial t} = \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)^2 + V \qquad \Rightarrow \qquad \boxed{i\hbar\frac{\partial|\Psi\rangle}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2|\Psi\rangle}{\partial x^2} + V|\Psi\rangle} \tag{1.59}$$

This above equation is called as the Time-Dependent Schrodinger's equation which is the quantum analog of the Energy conservation equation Total Energy $= K.E + P.E$.

1.2.5 Angular Momentum

Angular momentum is a vector given by, $\vec{L} = L_x\hat{i} + L_y\hat{j} + L_z\hat{k}$ whose magnitude square is $L^2 = L_x^2 + L_y^2 + L_z^2$. Employing the quantum analog, the relation between angular and linear momentum is $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$. Hence,

$$\hat{\mathbf{L}}_x = \hat{\mathbf{y}}\hat{\mathbf{p}}_z - \hat{\mathbf{z}}\hat{\mathbf{p}}_y \qquad \hat{\mathbf{L}}_y = \hat{\mathbf{z}}\hat{\mathbf{p}}_x - \hat{\mathbf{x}}\hat{\mathbf{p}}_z \qquad \hat{\mathbf{L}}_z = \hat{\mathbf{x}}\hat{\mathbf{p}}_y - \hat{\mathbf{y}}\hat{\mathbf{p}}_x \tag{1.60}$$

$$\Rightarrow \quad \hat{\mathbf{L}}_x = \frac{y}{i}\frac{\partial}{\partial z} - \frac{z}{i}\frac{\partial}{\partial y} \qquad \hat{\mathbf{L}}_y = \frac{z}{i}\frac{\partial}{\partial x} - \frac{x}{i}\frac{\partial}{\partial z} \qquad \hat{\mathbf{L}}_z = \frac{x}{i}\frac{\partial}{\partial y} - \frac{y}{i}\frac{\partial}{\partial x} \tag{1.61}$$

From these operators, we can get the commutator brackets of the components of Angular momentum as,

$$[\hat{\mathbf{L}}_x, \hat{\mathbf{L}}_y] = i\hat{\mathbf{L}}_z \qquad [\hat{\mathbf{L}}_y, \hat{\mathbf{L}}_z] = i\hat{\mathbf{L}}_x \qquad [\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_x] = i\hat{\mathbf{L}}_y \tag{1.62}$$

Since no component of Angular momentum commutes, we can say that no two components of Angular momentum can be found simultaneously. Consider the commutator relations,

$$[\hat{\mathbf{L}}^2, \hat{\mathbf{L}}_x] = 0 \qquad [\hat{\mathbf{L}}^2, \hat{\mathbf{L}}_y] = 0 \qquad [\hat{\mathbf{L}}^2, \hat{\mathbf{L}}_z] = 0 \tag{1.63}$$

These relations imply that we can measure the total magnitude of the angular momentum with only one of the components simultaneously and the other two will be unknown. So, let us arbitrarily choose to measure the magnitude $\hat{\mathbf{L}}$ and the $z-$component $\hat{\mathbf{L}}_z$ simultaneously. And since these two operators commute, by theorem(4) we can say that the two operators have the same eigen functions.

$$\Rightarrow \qquad \hat{\mathbf{L}}^2\Psi = \lambda_1\Psi \qquad \text{and} \qquad \hat{\mathbf{L}}_z\Psi = \lambda_2\Psi \tag{1.64}$$

Now let us use the concept of ladder operators. So let us define two operators,

$$\hat{\mathbf{L}}_+ = \hat{\mathbf{L}}_x + i\hat{\mathbf{L}}_y \qquad \text{and} \qquad \hat{\mathbf{L}}_- = \hat{\mathbf{L}}_x - i\hat{\mathbf{L}}_y \tag{1.65}$$

Since $\hat{\mathbf{L}}_x$ and $\hat{\mathbf{L}}_y$ commute with $\hat{\mathbf{L}}^2$ we can tell that even the ladder operators $\hat{\mathbf{L}}_\pm$ will commute with $\hat{\mathbf{L}}^2$. Thus even the ladder operators have the same Eigen function Ψ. The commutator of $\hat{\mathbf{L}}_z$ and $\hat{\mathbf{L}}_\pm$ is,

$$[\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_\pm] = [\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_x \pm i\hat{\mathbf{L}}_y] = [\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_x] \pm i[\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_y] = i\hat{\mathbf{L}}_y \pm i(-i\hat{\mathbf{L}}_x) = \pm(\hat{\mathbf{L}}_x \pm i\hat{\mathbf{L}}_y) = \pm\hat{\mathbf{L}}_\pm \tag{1.66}$$

$$\text{Now} \qquad \hat{\mathbf{L}}_z(\hat{\mathbf{L}}_\pm\Psi) = \hat{\mathbf{L}}_z\hat{\mathbf{L}}_\pm\Psi = \hat{\mathbf{L}}_z\hat{\mathbf{L}}_\pm\Psi - \hat{\mathbf{L}}_\pm\hat{\mathbf{L}}_z\Psi + \hat{\mathbf{L}}_\pm\hat{\mathbf{L}}_z\Psi = (\hat{\mathbf{L}}_z\hat{\mathbf{L}}_\pm - \hat{\mathbf{L}}_\pm\hat{\mathbf{L}}_z)\Psi + \hat{\mathbf{L}}_\pm\hat{\mathbf{L}}_z\Psi \tag{1.67}$$

$$= [\hat{\mathbf{L}}_z, \hat{\mathbf{L}}_\pm]\Psi + \hat{\mathbf{L}}_\pm\hat{\mathbf{L}}_z\Psi = \pm\hat{\mathbf{L}}_\pm\Psi + \hat{\mathbf{L}}_\pm(\lambda_2\Psi) = (\lambda_2 \pm 1)(\hat{\mathbf{L}}_\pm\Psi) \tag{1.68}$$

Thus we see that $\hat{\mathbf{L}}_\pm$ operator will add or subtract a factor of 1 to the Eigen value. Thus when the Eigen values are extremized, i.e., the ladder operator has to stop adding or removing the factor of 1 after the l^{th} energy states. Thus we get that $\lambda_1 = l(l+1)$ where $l = 0, 1, 2, \ldots$. Based on the l values we get $\lambda_2 = m$.

$$\Rightarrow \quad \boxed{\hat{\mathbf{L}}^2\Psi_l^m = l(l+1)\Psi_l^m \qquad \text{and} \qquad \hat{\mathbf{L}}_z\Psi_l^m = m_l\Psi_l^m} \tag{1.69}$$

The 'm_l' values have to be such that the separation between consecutive 'm_l' values is one and also it has to be symmetrical about the origin because of the rotational invariance of the quantum system. Thus the only possible values of 'm_l' are $-l, (-l+1), (-l+2), \ldots, 0, \ldots, (l-2), (l-1), l$. And for a given value of l there are $2l+1$ values of m_l. Thus the Eigen values of L^2 is $l(l+1)$, i.e., the magnitude of angular momentum can take values of only $\sqrt{2},\ \sqrt{6},\ \sqrt{12}, \ldots$. And moreover we see that the z-component of angular momentum can be only integer values, i.e., $L_z = 1,\ 2,\ 3, \ldots$.

We have to note that these results are valid for any 3D quantum system. So, any vector, say $\vec{J}$ that satisfies the Eq.(1.62) has the result of $J^2 = j(j+1)$ and $J_z = m_j$. There is also no condition that j has to be an integer, it can also be fractions as the result was got just by some algebraic method. And for a given j, m_j can take values of $-j, (-j+1), (-j+2), \ldots, 0, \ldots, (j-2), (j-1), j$.

1.2.6 Spin in NRQM

Angular momentum of the particle depends on its spatial coordinates which is analogous to the orbital angular momentum of classical bodies. But there is also another type of angular momentum associated with the particle even when there is no angular motion. For example, if there is a system where an electron is having linear motion only (orbital angular momentum is zero), still when measured there was a non-zero total angular momentum (Stern-Gerlach Experiment), which implies that the particle should have an intrinsic angular momentum. The particle's spin has nothing to do with the classical rotational motion of particles. But it is present intrinsically. Since spin angular momentum is a vector operator the exact same math is enough to work on spin angular momentum which was formulated for the Angular momentum in the previous section. Thus we can write,

$$\vec{S} = S_x\hat{i} + S_y\hat{j} + S_z\hat{k} \qquad \text{and} \qquad S^2 = S_x^2 + S_y^2 + S_z^2 \tag{1.70}$$

$$\Rightarrow \quad [\hat{\mathbf{S}}_x, \hat{\mathbf{S}}_y] = i\hat{\mathbf{S}}_z \qquad [\hat{\mathbf{S}}_y, \hat{\mathbf{S}}_z] = i\hat{\mathbf{S}}_x \qquad [\hat{\mathbf{S}}_z, \hat{\mathbf{S}}_x] = i\hat{\mathbf{S}}_y \tag{1.71}$$

$$\text{Also,} \quad [\hat{\mathbf{S}}^2, \hat{\mathbf{S}}_x] = 0 \qquad [\hat{\mathbf{S}}^2, \hat{\mathbf{S}}_y] = 0 \qquad [\hat{\mathbf{S}}^2, \hat{\mathbf{S}}_z] = 0 \tag{1.72}$$

Thus only the magnitude of the spin and one of its components (say S_z) can be measured. Hence we get,

$$\hat{\mathbf{S}}^2\Psi = s(s+1)\Psi \qquad \text{and} \qquad \hat{\mathbf{S}}_z\Psi = m_s\Psi \tag{1.73}$$

The Ladder operators are defined by, $\quad \hat{\mathbf{S}}_+ = \hat{\mathbf{S}}_x + i\hat{\mathbf{S}}_y \qquad \hat{\mathbf{S}}_- = \hat{\mathbf{S}}_x - i\hat{\mathbf{S}}_y \qquad (1.74)$

Here s is the spin quantum number and m_s is the magnetic spin quantum number where s to be half-integral or integral values and m_s can take values from, $-s, (-s+1), (-s+2), \ldots, 0, \ldots, (s-2), (s-1), s$ ..

Spin 1/2 Particles

They are the particles whose spin quantum number is given to be $s = 1/2$. These particles are of special interest as most of the particles we encounter are particles whose spin is half-integer. There are two Eigen states for spin state for such particles which are Spin-Up denoted by $|\uparrow\rangle = |1/2\rangle$ and for Spin-Down it is denoted by $|\downarrow\rangle = |-1/2\rangle$. So using these as basis vectors we can get the general state for the half-spin particle which can be expressed by a two-column matrix as,

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix} = a\hat{\chi}_+ + b\hat{\chi}_- \qquad \text{where} \qquad \underbrace{\hat{\chi}_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{spin-up}} \qquad \text{and} \qquad \underbrace{\hat{\chi}_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\text{spin down}} \tag{1.75}$$

So $\hat{\chi}_+$ and $\hat{\chi}_+$ are state vectors where the spin operators can operate on. Thus we get,

$$\hat{\mathbf{S}}^2\hat{\chi}_+ = s(s+1)\hat{\chi}_+ \qquad \text{and} \qquad \hat{\mathbf{S}}^2\hat{\chi}_- = s(s+1)\hat{\chi}_- \tag{1.76}$$

$$\text{Now as } s = 1/2 \text{ we get,} \qquad \hat{\mathbf{S}}^2\chi_x = \frac{3}{4}\hat{\chi}_+ \qquad \text{and} \qquad \hat{\mathbf{S}}^2\hat{\chi}_- = \frac{3}{4}\hat{\chi}_- \tag{1.77}$$

$$\Rightarrow \qquad \hat{\mathbf{S}}^2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad \hat{\mathbf{S}}^2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1.78}$$

So $\hat{\mathbf{S}}^2$ is a matrix operator which has 2 rows and 2 columns. And when worked out we get it to be,

$$\boxed{\hat{\mathbf{S}}^2 = \frac{3}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}} \tag{1.79}$$

Similarly, we can also find the $\hat{\mathbf{S}}_z$ operator which gives the $z-$component of spin angular momentum . Since the spin quantum number is $s = \frac{1}{2} \ \Rightarrow m_s = \pm\frac{1}{2}$ which corresponds to spin-up and spin-down.

$$\Rightarrow \qquad \hat{\mathbf{S}}_z\hat{\chi}_+ = \frac{1}{2}\hat{\chi}_+ \qquad \text{and} \qquad \hat{\mathbf{S}}_z\hat{\chi}_- = -\frac{1}{2}\hat{\chi}_- \tag{1.80}$$

When the expression of $\hat{\chi}_+$ and $\hat{\chi}_-$ is used and solved for $\hat{\mathbf{S}}_z$, we get its matrix form to be,

$$\boxed{\hat{\mathbf{S}}_z = \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}} \tag{1.81}$$

To get the $\hat{\mathbf{S}}_x$ and $\hat{\mathbf{S}}_y$ matrices we shall use the ladder operators which results in,

$$\hat{\mathbf{S}}_+\hat{\chi}_- = \hat{\chi}_+ \qquad \hat{\mathbf{S}}_-\hat{\chi}_+ = \hat{\chi}_- \qquad \hat{\mathbf{S}}_+\hat{\chi}_+ = \hat{\mathbf{S}}_-\hat{\chi}_- = 0 \tag{1.82}$$

Expanding the Ladder spin matrices in matrix notations and when solved for $\hat{\mathbf{S}}_+$ and $\hat{\mathbf{S}}_-$, we get,

$$\boxed{\hat{\mathbf{S}}_+ = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} \qquad \text{and} \qquad \hat{\mathbf{S}}_- = \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}} \tag{1.83}$$

Now to get $\hat{\mathbf{S}}_x$ and $\hat{\mathbf{S}}_y$ we can use,

$$\hat{\mathbf{S}}_x = \frac{\hat{\mathbf{S}}_+ + \hat{\mathbf{S}}_-}{2} \qquad \text{and} \qquad \hat{\mathbf{S}}_y = \frac{\hat{\mathbf{S}}_+ - \hat{\mathbf{S}}_-}{2i} \tag{1.84}$$

Thus we get the matrix form for $\hat{\mathbf{S}}_x$ and $\hat{\mathbf{S}}_y$ as,

$$\boxed{\hat{\mathbf{S}}_x = \frac{1}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \qquad \text{and} \qquad \hat{\mathbf{S}}_y = \frac{1}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}} \tag{1.85}$$

As all these operators $\hat{\mathbf{S}}_x$, $\hat{\mathbf{S}}_y$, $\hat{\mathbf{S}}_z$ have a factor of $1/2$. So they can be combined and written as,

$$\hat{\mathbf{S}} = \frac{\hat{\sigma}}{2} \tag{1.86}$$

where $\hat{\sigma} = \hat{\sigma}_x\hat{i} + \hat{\sigma}_y\hat{s} + \hat{\sigma}_z\hat{k}$ which are vector operators called as *Pauli Spin matrices* given by,

$$\boxed{\hat{\sigma}_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \qquad \hat{\sigma}_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} \qquad \hat{\sigma}_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}} \tag{1.87}$$

We can see that the Spin matrices $\hat{\mathbf{S}}^2$, $\hat{\mathbf{S}}_x$, $\hat{\mathbf{S}}_y$, $\hat{\mathbf{S}}_z$ are all Hermitian matrices but $\hat{\mathbf{S}}_+$, $\hat{\mathbf{S}}_-$ are not Hermitian matrices. Thus only $\hat{\mathbf{S}}^2$, $\hat{\mathbf{S}}_x$, $\hat{\mathbf{S}}_y$, $\hat{\mathbf{S}}_z$ are observables and can be measured as their eigen values and expectation values are real. But for $\hat{\mathbf{S}}_+$, $\hat{\mathbf{S}}_-$ as they are not hermitian they don't have real observables.

Now if we try to measure the spin of the particle in some general state χ where $\chi = a\hat{\chi}_+ + b\hat{\chi}_-$ then the probability of measuring the spin to be $+1/2$ i.e., with the state $\hat{\chi}_+$ is $|a|^2$ and the probability of measuring the spin to be $-1/2$ i.e., with the state $\hat{\chi}_-$ is $|b|^2$. Since these are the only two possible states that can be observed, the total probability $|a|^2 + |b|^2 = 1$ which is the normalization condition for the spinor χ.

We have to note that the spin is a new degree of freedom that a particle can occupy. Since $s = 1/2$ we get $m_s = \pm 1/2$ which implies that particle along with having the quantum numbers n, l, m can have either spin-up state and spin down state. Thus there are two types of wave functions for a particle given by Ψ_+ for a particle with spin-up state and Ψ_- for a particle with spin down state. Where m_l is the orbital magnetic quantum number associated with the extrinsic angular momentum and m_s is the spin magnetic quantum number associated with the intrinsic spin of the particle.

Half Integral spin ($n/2$) and Integral spin (n) Particles

Broadly all the particles can be classified into two types.

1. Particles with Integer-spins are called Bosons whose behavior is governed by a particular class of quantum statistics called the Bose-Einstein statistics.
2. Particles with Half-integer spins are called Fermions whose behavior is governed by a class of quantum statistics called the Fermi-Dirac Statistics.

Consider a two-particle system whose individual wave function is $\Psi(\vec{r}_a, t)$ and $\Psi(\vec{r}_b, t)$. The total wave function of the system is given by $\Psi(\vec{r}_a, \vec{r}_b, t)$. So one single wave function describes the two particles. Then can we distinguish between the states of two particles? i.e., could we tell that specific particle is in state Ψ_a and other is in Ψ_b?? No, because once the particles are together they lose their individuality. We cannot name an electron and keep track of it because they are not observable unless we change the particle's state. Now the Hamiltonian of the system is given by,

$$\hat{\mathbf{H}} = -\frac{\hbar^2}{2m_a}\nabla_a^2 - \frac{\hbar^2}{2m_b}\nabla_b^2 + V \tag{1.88}$$

The subscripts indicate that the differentiation is done with respect to their respective coordinates. Now the wave function can be separable if the potential energy is time-independent. Then we get,

$$\Psi(\vec{r}_a, \vec{r}_b, t) = \psi(\vec{r}_a, \vec{r}_b)e^{i\frac{E}{\hbar}t} \tag{1.89}$$

The time factor is the same as E is the total energy of the system and not of the individual particles. Now the wave function of the system can be either $\psi(\vec{r}_a, \vec{r}_b)$ or $\psi(\vec{r}_b, \vec{r}_a)$. Then the total wave function is given by the linear combination,

$$\Psi = c_1\psi(\vec{r}_a, \vec{r}_b) \pm c_2\psi(\vec{r}_b, \vec{r}_a) \tag{1.90}$$

When the normalization condition is applied we get $c_1 = c_2 = \frac{1}{\sqrt{2}}$, thus,

$$\Psi = \frac{1}{\sqrt{2}}\Big[\psi(\vec{r}_a, \vec{r}_b) \pm \psi(\vec{r}_b, \vec{r}_a)\Big] \tag{1.91}$$

It is seen that for particles with integer spins, their wave functions are anti-symmetric and for particles with integer spins, their wave function is symmetric, i.e.,

$$\psi(\vec{r}_a, \vec{r}_b) = +\psi(\vec{r}_b, \vec{r}_a) \qquad \Rightarrow \qquad \text{Boson} \tag{1.92}$$

$$\psi(\vec{r}_a, \vec{r}_b) = -\psi(\vec{r}_b, \vec{r}_a) \qquad \Rightarrow \qquad \text{Fermion} \tag{1.93}$$

$$\text{Boson} \qquad \Rightarrow \qquad \Psi_{\text{Boson}} = \frac{1}{\sqrt{2}}\Big[\psi(\vec{r}_a, \vec{r}_b) + \psi(\vec{r}_b, \vec{r}_a)\Big] \tag{1.94}$$

$$\text{Fermion} \qquad \Rightarrow \qquad \Psi_{\text{Fermion}} = \frac{1}{\sqrt{2}}\Big[\psi(\vec{r}_a, \vec{r}_b) - \psi(\vec{r}_b, \vec{r}_a)\Big] \tag{1.95}$$

If all the quantum numbers are the same for both the particles, i.e., if both the particles occupy the same energy state, then if $\psi(\vec{r}_a, \vec{r}_b) = \psi(\vec{r}, \vec{r})$ we get,

$$\text{Boson} \qquad \Rightarrow \qquad \Psi_{\text{Boson}} = \sqrt{2}\psi^2 \qquad \Rightarrow \qquad |\Psi_{\text{Boson}}| = 2|\psi| \tag{1.96}$$

$$\text{Fermion} \qquad \Rightarrow \qquad \Psi_{\text{Fermion}} = 0 \qquad \Rightarrow \qquad |\Psi_{\text{Fermion}}| = 0 \tag{1.97}$$

Thus for Bosons, if two particles with the same quantum state are in the same system they become one single entity whose probability density function doubles, i.e., bosons are more likely to occupy the same quantum state. And for Fermion the probability density becomes zero, i.e., the probability of fermions being in the same energy state is Zero. So no two fermions in the same system can occupy the same energy state. This is known as the '*Pauli's exclusion principle*'.

Chapter 2

1^{st} attempt for a Relativistic wave equation

2.1 Problems with Non-Relativistic Quantum Mechanics

1. Non invariance of Schrodinger's equation under Lorentz Transformation

The Schrodinger equation that governs the quantum realm is invariant under Galilean transformation (Eq.1.1) which is appropriate at low speeds ($v << c$). But for particles or systems at speeds closer to that of light, Galilean transformation breaks down and Lorentz transformations are better approximation. So then if the Schrodinger equation describes fast moving particles then it has to be invariant under Lorentz transformation. So let us use Lorentz transformation Eq.(1.4) on the Schrodinger equation.

$$\underbrace{i\frac{\partial \Psi}{\partial t} = -\frac{1}{2m}\frac{\partial^2 \Psi}{\partial x^2} + V\Psi}_{\text{Schrodinger equation}} \qquad \text{and} \qquad \underbrace{x' = \gamma\,(x - vt) \qquad\qquad t' = \gamma\,(t - xv)}_{\text{Lorentz Transformations}} \tag{2.1}$$

$$\frac{\partial \Psi}{\partial t} = \frac{\partial \Psi}{\partial t'}\frac{\partial t'}{\partial t} + \frac{\partial \Psi}{\partial x'}\frac{\partial x'}{\partial t} = \frac{\partial \Psi}{\partial t'}\frac{\partial}{\partial t}\big(\gamma(t - xv)\big) + \frac{\partial \Psi}{\partial x'}\frac{\partial}{\partial t}\big(\gamma(x - vt)\big) = \gamma\frac{\partial \Psi}{\partial t'} - \gamma v\frac{\partial \Psi}{\partial x'} \tag{2.2}$$

$$\frac{\partial \Psi}{\partial x} = \frac{\partial \Psi}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial \Psi}{\partial t'}\frac{\partial t'}{\partial x} = \frac{\partial \Psi}{\partial x'}\frac{\partial}{\partial x}\big(\gamma(x - vt)\big) + \frac{\partial \Psi}{\partial t'}\frac{\partial}{\partial x}\big(\gamma(t - vx)\big) = \gamma\frac{\partial \Psi}{\partial x'} - \gamma v\frac{\partial \Psi}{\partial t'} \tag{2.3}$$

$$\Rightarrow \qquad \frac{\partial^2 \Psi}{\partial x^2} = \frac{\partial}{\partial x}\Big(\frac{\partial \Psi}{\partial x}\Big) = \frac{\partial}{\partial x}\Big(\gamma\frac{\partial \Psi}{\partial x'} - \gamma v\frac{\partial \Psi}{\partial t'}\Big) = \gamma^2\frac{\partial^2 \Psi}{\partial x'^2} - 2v\gamma^2\frac{\partial^2 \Psi}{\partial x'\partial t'} + \gamma^2 v^2\frac{\partial^2 \Psi}{\partial t'^2} \tag{2.4}$$

Substituting the primed derivatives in the Schrodinger equation we get,

$$i\gamma\Big(\frac{\partial \Psi}{\partial t'} - v\frac{\partial \Psi}{\partial x'}\Big) = -\frac{\gamma^2}{2m}\Big(\frac{\partial^2 \Psi}{\partial x'^2} - 2v\frac{\partial^2 \Psi}{\partial x'\partial t'} + v^2\frac{\partial^2 \Psi}{\partial t'^2}\Big) + V\Psi \tag{2.5}$$

This is the Lorentz transformed Schrodinger equation. Clearly, we can see that the form of the equation is completely changed. Thus we can conclude that the Schrodinger equation is not a Lorentz invariant equation. The transformed equation is not a linear partial differential equation anymore because of the double derivative term of Ψ with respect to both x' and t'. So, Schrodinger equation is invalid at relativistic speeds.

Moreover Schrodinger's equation is an Energy Eigen value equation when potential energy is time independent. So in some sense, Schrodinger's equation gives information only about energy levels of a system and its quantization along with the conservation of probability and nothing more.

2. Unable to explain why Intrinsic Spin exists

Non-relativistic quantum mechanics (NRQM) and Schrodinger's equation was not able to explain why intrinsic spin angular momentum of the particle exists. Schrodinger's equation was not able to deduce the Spin quantum number. And treats both up spin and down spin particles the same way. The equations Eq.(1.92) and Eq.(1.93) were not explained by NRQM basis rather it was got by the relativistic methods. Where the spin angular momentum invariably arise to conserve the total angular momentum of a system.

2.2 Klein-Gordon Equation

According to special relativity, the total energy of a particle of mass 'm' in motion is given by $E^2 = m^2 + p^2$. The corresponding operators of total energy and momentum instead of the observables are,

$$\hat{\mathbf{P}} = -i\vec{\nabla} \quad \Rightarrow \quad \hat{\mathbf{P}}^2 = \hat{\mathbf{P}} \cdot \hat{\mathbf{P}} = -\nabla^2 \qquad \text{and} \qquad \hat{\mathbf{E}} = i\frac{\partial}{\partial t} \quad \Rightarrow \quad \hat{\mathbf{E}}^2 = \hat{\mathbf{E}} \cdot \hat{\mathbf{E}} = -\frac{\partial^2}{\partial t^2} \tag{2.6}$$

By quantizing the mass-energy equivalence relation using these operators we get, $\dfrac{\partial^2}{\partial t^2} - \nabla^2 + m^2 = 0$. The operator $\dfrac{\partial}{\partial t^2} - \nabla^2$ is the d'alembertian operator (Eq.(1.30)). Operating the above equation with a wave function ϕ we get a relativistic wave equation called as the Klein Gordan (K.G.) equation as,

$$\boxed{\frac{\partial^2\phi}{\partial t^2} - \frac{\partial^2\phi}{\partial x^2} - \frac{\partial^2\phi}{\partial y^2} - \frac{\partial^2\phi}{\partial z^2} + m^2\phi = 0} \quad \text{or} \quad \boxed{(\Box + m^2)\phi = 0} \quad \text{or} \quad \boxed{(\partial_\mu\partial^\mu + m^2)\phi = 0} \tag{2.7}$$

This relativistic equation is no longer a linear equation in time. And has two solutions for energy given by

$$E = \pm\sqrt{p^2 + m^2} \tag{2.8}$$

Thus Klein Gordon equation predicts particles of the same mass and opposite energies (positive and negative). Thus energy is no more positive definite. And the significance of particles with negative energies will be seen in Chapter(4). The particle described here is a free particle because there is no term involving the potential energy of the system. Now if the particle is massless, then $m = 0$ this implies

$$\Box\phi = 0 \quad \text{or} \quad \frac{\partial^2\phi}{\partial t^2} = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} \quad \text{(in SI units for 1D we get)} \quad \frac{\partial^2\phi}{\partial t^2} = c^2\frac{\partial^2\phi}{\partial x^2} \tag{2.9}$$

This is the wave equation that describes waves. Hence waves generally are described by K.G equation.

2.3 Relativistic Complex field and the description of charge

The Lagrangian of spin-zero Scalar particles is given by a scalar complex field ϕ and its conjugate ϕ^* as,

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi^* - \frac{1}{2}m^2(\phi\phi^*) \tag{2.10}$$

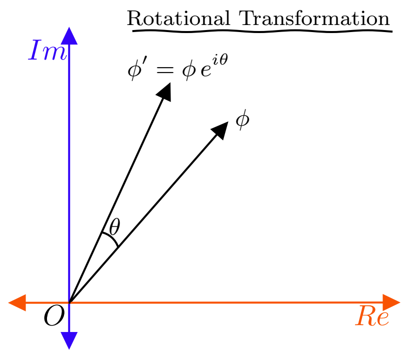
Whose equations of motion are given by the Euler Lagrange equations of motion as,

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0 \qquad \text{and} \qquad \frac{\partial\mathcal{L}}{\partial\phi^*} - \partial^\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial^\mu\phi^*)}\right) = 0 \tag{2.11}$$

Using this we get, $$-\frac{1}{2}m^2\phi^* - \frac{1}{2}\partial_\mu\partial^\mu\phi^* = 0 \quad \text{and} \quad -\frac{1}{2}m^2\phi - \frac{1}{2}\partial^\mu\partial_\mu\phi = 0 \tag{2.12}$$

Which results in $$(\partial_\mu\partial^\mu + m^2)\phi^* = 0 \quad \text{and} \quad (\partial^\mu\partial_\mu + m^2)\phi = 0 \quad \rightarrow \quad \text{K.G. Equation} \tag{2.13}$$

Thus the Lagrangian which we chose is indeed for the spin-zero free particles. So now let us do a rotational transformation of this scalar field. So if we have to rotate (transform) the scalar field ϕ then we have to multiply it by the rotation factor $e^{i\theta}$ which rotates the complex vector by an angle of θ.



Rotational transformation is given by, $\qquad \phi' \;\rightarrow\; \phi\, e^{i\theta} \quad$ and $\quad \phi'^{*} \;\rightarrow\; \phi^{*} e^{-i\theta} \qquad (2.14)$

$$\Rightarrow \quad \partial_\mu \phi' = (\partial_\mu \phi) e^{i\theta} \quad \text{and} \quad \partial_\mu \phi'^{*} = (\partial_\mu \phi^{*}) e^{-i\theta} \tag{2.15}$$

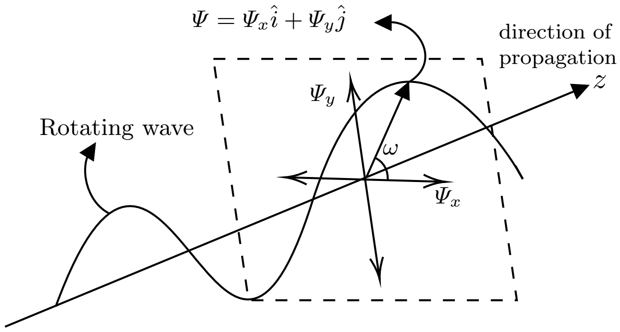
$$\Rightarrow \quad \phi' \, \phi'^{*} = \phi \cancel{e^{i\theta}} \cdot \phi^{*} \cancel{e^{-i\theta}} = \phi\, \phi^{*} \quad \text{and} \quad \partial_\mu \phi' \, \partial^\mu \phi'^{*} = (\partial_\mu \phi)\cancel{e^{i\theta}} \cdot (\partial_\mu \phi^{*})\cancel{e^{-i\theta}} = \partial_\mu \phi \, \partial^\mu \phi^{*} \tag{2.16}$$

Thus we see that the terms like $\phi\phi^{*}$ and $\partial_\mu \phi \partial^\mu \phi^{*}$ are invariant under rotational transformations. And since the Lagrangian density contains only them, it is invariant under complex rotation. Noether's theorem states that 'For every symmetry in nature, there is an associated conserved quantity.' So because of the symmetry in the Lagrangian under rotational transformation, there has to be a conserved quantity associated with it. Since the complex field has a real and an imaginary part, the conserved quantity given by,

$$\int \Big(\Pi_\mu (\delta\phi) + \Pi^{*}_\mu (\delta\phi^{*}) \Big) d^4x = \int \Big(\frac{\partial \mathcal{L}}{\partial \phi_\mu} (\delta\phi) + \frac{\partial \mathcal{L}}{\partial \phi^{*}_\mu} (\delta\phi^{*}) \Big) d^4x = \text{constant} \tag{2.17}$$

$$\Rightarrow \quad \frac{i}{2} \int \left(\phi \frac{\partial \phi^{*}}{\partial x_\mu} - \phi^{*} \frac{\partial \phi}{\partial x_\mu} \right) d^4x = \text{constant} \tag{2.18}$$

When the same math is done in a normal three-dimensional space where for a wave propagating in a particular direction ($z-$axis) where particles oscillate in two perpendicular directions ($x = \phi_x$ and $y = \phi_y$) as shown in the below figure, then the conserved quantity becomes the angular momentum per unit length of the wave given by $L_z = x\dot{y} - \dot{x}y$ of the wave whose axis is along the direction of propagation where the angular momentum travels along with the direction of propagation of wave as $L_z \rightarrow z$-component.



Thus the conserved quantity represents the charge of the quantum of the field where the charged particles are represented by complex fields. Since the quantity is having space and time derivatives, in some sense the complex charge vector rotates as time proceeds and also as the charged particle moves in space. Thus charge is like an analog of the angular momentum of the rotating complex charge vector.

The quantity given by, $\quad \dfrac{i}{2} \displaystyle\int \left(\phi \frac{\partial \phi^{*}}{\partial x^\mu} - \phi^{*} \frac{\partial \phi}{\partial x^\mu} \right) d^4x = j^\mu \quad$ is called the four-current. $\qquad (2.19)$

The four current j^μ has the components of (ρ, j_x, j_y, j_z) where the time component $j^0 = \rho$ is the charge density and spatial components $\vec{j} = (j_x, j_y, j_z)$ is the current density. Since j^μ is a constant the four gradient of j^μ is zero. Hence we get,

$$\partial_\mu j^\mu = 0 \qquad \Rightarrow \qquad \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0 \qquad \text{(Equation of conservation of charge)} \tag{2.20}$$

Thus the charge density ρ is got by the time component of j^μ which is, $\qquad \rho = \phi \dfrac{\partial \phi^{*}}{\partial t} - \phi^{*} \dfrac{\partial \phi}{\partial t} \qquad (2.21)$

For stationary states solution, i.e., $\phi = \psi(\vec{r}) e^{-iEt}$ we get the charge density to be $\rho = \frac{E}{m} \psi^{*} \psi$. This is similar to the NRQM limit but there is a factor of $\frac{E}{m}$ present. This is because the three dimensional volume element gets Lorentz contracted which corresponds to the increase in the density by the factor of $\frac{E}{m}$. Now since E can be positive or negative we get ρ also to be positive or negative. If ϕ is a real field then $\phi = \phi^{*}$ which implies that the charge of the particle is $\rho = 0$. Then that describes neutral particles with all kinds of charge to be zero which can be created and annihilated singly. For example, a photon is a neutral particle that can be emitted or absorbed as a single particle during the transition of a system in energy states.

Chapter 3

The Dirac Theory

3.1 Dirac Equation

Klein Gordon equation best describes only scalar free particles. For other types of particles with non-zero spin, it doesn't work. This limitation was mainly because of the second-order differential equation that we come across in $E^2 = P^2 + m^2$. So to come out of that limitation we need to brute force our equation to be a first-order linear equation. So let us bring the four-vector notation first. We know that four-momentum is given by, $P^\mu = (P^0, P^1, P^2, P^3) = (E, P_x, P_y, P_z)$ and its dual is $P_\mu = (P_0, P_1, P_2, P_3) = (E, -P_x, -P_y, -P_z)$. Thus we get, $P_\mu P^\mu = E^2 - P_x^2 - P_y^2 - P_z^2 = E^2 - P^2$. But according to mass-energy equivalence we get $E^2 - P^2 = m^2$. Thus in four-vector notation, mass-energy equivalence is given by,

$$P_\mu P^\mu = m^2 \qquad \text{or} \qquad P_\mu P^\mu - m^2 = 0 \tag{3.1}$$

The above equation should be made first order by factorization. Consider the below factorized equation,

$$(P_\mu + m)(P^\mu - m) = 0 \quad ✗ \quad \text{This is dimensionally incorrect.} \tag{3.2}$$

Since 'm' is a scalar we shall try multiplying the Four momentum with some four-vector (say γ^μ and γ^ν) so that it becomes a scalar. Hence we can write,

$$(\gamma^\nu P_\nu + m)(\gamma^\mu P_\mu - m) = 0 \tag{3.3}$$

Now we shall impose some restrictions on this new four-vector so that when the LHS is expanded we need it to yield back Eq.(3.1). So the above equation when expanded becomes,

$$\gamma^\nu \gamma^\mu P_\nu P_\mu - m^2 + (\gamma^\mu P_\mu - \gamma^\nu P_\nu)m = 0 \tag{3.4}$$

Since the cross term $(\gamma^\mu P_\mu - \gamma^\nu P_\nu)m$ is not needed we can put a condition to the '*gamma*' vector as,

$$\text{First Condition} \quad \rightarrow \quad \gamma^\mu P_\mu - \gamma^\nu P_\nu = 0 \qquad \text{or} \qquad \gamma^\mu P_\mu = \gamma^\nu P_\nu$$

This is obvious as μ and ν are just indices that run from $0, 1, 2, 3$. Then what we are left is,

$$\gamma^\nu \gamma^\mu P_\nu P_\mu - m^2 = 0 \tag{3.5}$$

Expanding this,
$$\begin{aligned}\gamma^\nu \gamma^\mu P_\nu P_\mu - m^2 = \; & \gamma^0\gamma^0 P_0P_0 + \gamma^0\gamma^1 P_0P_1 + \gamma^0\gamma^2 P_0P_2 + \gamma^0\gamma^3 P_0P_3 + \\ & \gamma^1\gamma^0 P_1P_0 + \gamma^1\gamma^1 P_1P_1 + \gamma^1\gamma^2 P_1P_2 + \gamma^1\gamma^3 P_1P_3 + \\ & \gamma^2\gamma^0 P_2P_0 + \gamma^2\gamma^1 P_2P_1 + \gamma^2\gamma^2 P_2P_2 + \gamma^2\gamma^3 P_2P_3 + \\ & \gamma^3\gamma^0 P_3P_0 + \gamma^3\gamma^1 P_3P_1 + \gamma^3\gamma^2 P_3P_2 + \gamma^3\gamma^3 P_3P_3 - m^2 = 0\end{aligned}$$

Now we shall group the diagonal elements whose indices are all same. Also notice that the momentum components are numbers so $P_\mu P_\nu = P_\nu P_\mu$, thus grouping such terms we get,

$$\gamma^\nu \gamma^\mu P_\nu P_\mu - m^2 = \sum_{\mu=\nu} \gamma^\nu \gamma^\mu P_\nu P_\mu + \sum_{\mu \neq \nu} P_\nu P_\mu (\gamma^\nu \gamma^\mu + \gamma^\mu \gamma^\nu) - m^2 = 0 \tag{3.6}$$

Thus we need the off-diagonal terms to become zero, so, we get another condition that,

$$\text{Second Condition} \quad \rightarrow \quad \gamma^\nu\gamma^\mu + \gamma^\mu\gamma^\nu = 0 \qquad (\text{for} \quad \mu \neq \nu) \tag{3.7}$$

Thus finally what we are left with is,
$$\sum_{\mu=\nu} \gamma^\nu\gamma^\mu P_\nu P_\mu - m^2 = 0 \tag{3.8}$$

Now we need this equation to look like Eq.(3.1). Thus the third condition is,

$$\gamma^\nu\gamma^\mu P_\nu P_\mu = P_\mu P^\mu \qquad \Rightarrow \qquad \gamma^\nu\gamma^\mu = \eta^{\mu\nu} \qquad (\text{Since,} \quad \eta^{\mu\nu}P_\nu = P^\mu \quad \text{by} \quad \text{Eq.}(1.6)) \tag{3.9}$$

Now the second and the third conditions go hand in hand as for metric tensor $\eta^{\mu\nu} = 0$ when $\mu \neq \nu$, i.e., off-diagonal terms are zero (by Eq.(1.5)). Also notice that $\gamma^\nu\gamma^\mu + \gamma^\mu\gamma^\nu = \{\gamma^\nu, \gamma^\mu\}$ is the anti-commutator bracket. Thus we can mix these two conditions as,

$$\{\gamma^\nu, \gamma^\mu\} = 2\eta^{\mu\nu} \tag{3.10}$$

Now any tensors that follow the above conditions is said to define 'Clifford algebra'. Borrowing the results of this algebra, we get the *gamma* matrices to be,

$$\gamma^0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \gamma^1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

$$\gamma^2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} \qquad \gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

If we closely observe the elements of *gamma* matrices can be neatly written as $\gamma^\mu = (\gamma^0, \gamma^i)$ where,

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \gamma^i = \begin{pmatrix} 0 & \hat{\sigma}^i \\ -\hat{\sigma}^i & 0 \end{pmatrix} \tag{3.11}$$

Here, 1 is the 2×2 identity matrix, 0 is the 2×2 zero matric and $\hat{\sigma}^i$ are the respective Pauli spin matrices as seen in Eq.(1.87). Thus we can quantize the four-vector representation of mass-energy equivalence by using the operator definition of total energy $\hat{\mathbf{E}}$, and momentum $\hat{\mathbf{P}}$.

$$\hat{\mathbf{E}} = i\frac{\partial}{\partial t} \qquad \hat{\mathbf{P}} = -i\vec{\nabla} = -i\Big(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\Big) \tag{3.12}$$

$$\Rightarrow \qquad P_\mu = (E, -\vec{P}) = i\Big(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big) = i\partial_\mu \tag{3.13}$$

Then using Eq.(3.3) we get,
$$(\gamma^\mu P_\mu - m) = 0 \qquad \Rightarrow \qquad (i\gamma^\mu\partial_\mu - m) = 0 \tag{3.14}$$

Using the Feynman's *slash notation*, i.e., for any four-vector a_μ, the quantity $\gamma^\mu a_\mu$ can be written as $\not{a}$. Also notice that the *LHS* of the above equation is an operator, so it makes sense only if it is acting on a function. Thus multiplying the operator with ψ we get the Dirac's Equation given by,

$$\boxed{(i\not{\partial} - m)\psi = 0} \tag{3.15}$$

This is a completely relativistic equation that should work for particles in even relativistic speeds. Now the term $\gamma^\mu\partial_\mu$ though appears to be a Lorentz scalar since γ^μ is a 4×4 tensor, the term $\gamma^\mu\partial_\mu$ is actually a 4×1 four-vector. Thus we need to multiply m with an identity matrix I_4 to keep the equation dimensionally consistent. Then in SI units, we get the Dirac equation to be,

$$\boxed{(i\hbar\not{\partial} - mcI_4)\psi = 0} \tag{3.16}$$

The function ψ which we have multiplied to the operator $i\partial\!\!\!/-m$ in the Dirac equation is a 4×1 matrix called as spinor (reason in section(3.2.3)) because only then it can be multiplied to the 4×4 gamma matrices.

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} \tag{3.17}$$

Hermitian adjoint of Dirac's equation

The Dirac's equation is given by $(i\partial\!\!\!/ - m)\psi = 0$ or $i\gamma^\mu\partial_\mu\psi - m\psi = 0$. First, let's find the conjugates of gamma matrices. Using the dagger operation given by $\dagger$ = Transpose conjugate, we get,

$$(\gamma^0)^\dagger = \gamma^0 \qquad (\gamma^i)^\dagger = -\gamma^i \qquad \Rightarrow \quad \gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0 \tag{3.18}$$

Let us take the hermitian conjugate of the Dirac equation $\big[(i\gamma^\mu\partial_\mu - m)\psi\big]^\dagger = 0$,

$$\Rightarrow \qquad \psi^\dagger\big(-i\overleftarrow{\partial}_\mu\gamma^{\mu\dagger} - m\big) = 0 \tag{3.19}$$

$$\Rightarrow \qquad \psi^\dagger\big(-i\overleftarrow{\partial}_\mu(\gamma^0\gamma^\mu\gamma^0) - m\big) = 0 \qquad \text{(using } \gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0\text{)} \tag{3.20}$$

$$\Rightarrow \qquad \psi^\dagger\gamma^0\big(-i\overleftarrow{\partial}_\mu(\gamma^\mu\gamma^0) - (\gamma^0)^{-1}m\big) = 0 \qquad \text{(taking } \gamma^0 \text{ common on left)} \tag{3.21}$$

$$\Rightarrow \qquad \psi^\dagger\gamma^0\big(-i\overleftarrow{\partial}_\mu(\gamma^\mu\gamma^0\gamma^0) - (\gamma^0)^{-1}m\gamma^0\big) = 0 \qquad \text{(multiplying } \gamma^0 \text{ from right)} \tag{3.22}$$

By Clifford's Algebra in Eq.(3.10) we get $\gamma^0\gamma^0 = \eta^{00}I = 1\cdot I = I$ and $(\gamma^0)^{-1}m\gamma^0 = (\gamma^0)^{-1}\gamma^0 m = Im$.

Thus the conjugate equation becomes, $$\psi^\dagger\gamma^0\big(-i\overleftarrow{\partial}_\mu\gamma^\mu - m\big) = 0 \tag{3.23}$$

Now using $\psi^\dagger\gamma^0 = \overline{\psi}$ and $\overleftarrow{\partial}_\mu\gamma^\mu = \overleftarrow{\partial\!\!\!/}$ and by taking the $-ve$ sign out we get the Dirac adjoint equation,

$$\boxed{\overline{\psi}(i\overleftarrow{\partial\!\!\!/} + m) = 0} \tag{3.24}$$

The reverse arrowhead in $\overleftarrow{\partial\!\!\!/}$ symbol is to indicate that the derivative acts on the left side i.e., on $\overline{\psi}$.

The Alpha and Beta matrices

The alpha and beta matrices are some important matrices that come in handy to denote certain quantities and often appear in relativistic equations. The alpha matrices denoted by $\vec{\alpha}$ has three component matrices which are $\vec{\alpha} = (\alpha_1, \alpha_2, \alpha_3) = \alpha_i$. They are related to the gamma matrices as,

$$\beta = \gamma^0 \qquad \text{and} \qquad \vec{\alpha} = (\alpha_1, \alpha_2, \alpha_3) = \alpha_i = \beta\gamma^i \tag{3.25}$$

$$\Rightarrow \qquad \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \text{and} \qquad \alpha_i = \begin{pmatrix} 0 & \hat{\sigma}_i \\ \hat{\sigma}_i & 0 \end{pmatrix} \qquad (\hat{\sigma} \quad \rightarrow \quad \text{Pauli matrices}) \tag{3.26}$$

By this notation the total energy operator for a free particle, i.e., its Hamiltonian operator is given by,

$$\hat{\mathbf{H}} = \vec{\alpha}\cdot\vec{P} + \beta m = \vec{\alpha}\cdot(i\vec{\nabla}) + \beta m \tag{3.27}$$

We can define a four-vector of the alpha matrices as α_μ whose time component is β and space components are α_i. Thus we get, $\alpha_\mu = (\beta, \beta\alpha_i)$. Because of the condition in Eq.(3.10) we get a relation among the different components of α matrices as,

$$\alpha_\nu\alpha_\mu + \alpha_\mu\alpha_\nu = \{\alpha_\nu, \alpha_\mu\} = 2\delta_{\mu\nu}I \tag{3.28a}$$

$$\alpha_\mu \text{ is a Hermitian matrix and } \alpha_\mu^2 = 1 \qquad \Rightarrow \quad \text{Eigen values of} \quad \alpha_\mu = \pm 1 \tag{3.28b}$$

3.2 Some Important results of Dirac Equation

3.2.1 Probability Density and Continuity Equation

When the Dirac equation is expanded, it is of the form, $$i\Big(\gamma^0\frac{\partial\psi}{\partial t}+\vec{\gamma}\cdot(\vec{\nabla}\psi)\Big)-m\psi=0 \tag{3.29}$$

Multiplying throughout by γ^0 on the left side we get, $$i\Big(\underbrace{(\gamma^0)^2}_{1}\frac{\partial\psi}{\partial t}+\underbrace{\gamma^0\vec{\gamma}}_{\vec{\alpha}}\cdot(\vec{\nabla}\psi)\Big)-\gamma^0m\psi=0$$

$$\Rightarrow\quad i\Big(\frac{\partial\psi}{\partial t}+\vec{\alpha}\cdot(\vec{\nabla}\psi)\Big)-\gamma^0m\psi=0 \tag{3.30}$$

Doing the same for the adjoint of dirac equation in Eq.(3.23) which is, $\psi^\dagger\gamma^0\big(i\overleftarrow{\partial}_\mu\gamma^\mu+m\big)=0$ we get,

$$i\psi^\dagger\gamma^0\Big(\frac{\overleftarrow{\partial}}{\partial t}\gamma^0+\overleftarrow{\nabla}\cdot\vec{\gamma}\Big)+\psi^\dagger\gamma^0m=0 \qquad\Rightarrow\qquad i\psi^\dagger\Big(\frac{\overleftarrow{\partial}}{\partial t}\underbrace{(\gamma^0)^2}_{1}+\overleftarrow{\nabla}\cdot\underbrace{(\gamma^0\vec{\gamma})}_{\vec{\alpha}}\Big)+\psi^\dagger\gamma^0m=0$$

$$\Rightarrow\qquad i\Big(\frac{\partial\psi^\dagger}{\partial t}+(\vec{\nabla}\psi^\dagger)\cdot\vec{\alpha}\Big)+\psi^\dagger\gamma^0m=0 \tag{3.31}$$

We shall try to eliminate the term independent of derivatives, i.e., the mass term. Doing the operation,

$$\psi^\dagger\times\text{Eq.(3.30)}+\text{Eq.(3.31)}\times\psi$$

$$\left(i\Big(\psi^\dagger\frac{\partial\psi}{\partial t}+\psi^\dagger\vec{\alpha}\cdot(\vec{\nabla}\psi)\Big)-\cancel{\psi^\dagger\gamma^0m\psi}=0\right)+\left(i\Big(\frac{\partial\psi^\dagger}{\partial t}\psi+(\vec{\nabla}\psi^\dagger)\cdot\vec{\alpha}\psi\Big)+\cancel{\psi^\dagger\gamma^0m\psi}\right)=0 \tag{3.32}$$

$$\Rightarrow\quad\Big(\psi^\dagger\frac{\partial\psi}{\partial t}+\frac{\partial\psi^\dagger}{\partial t}\psi\Big)+\Big(\psi^\dagger\vec{\alpha}\cdot(\vec{\nabla}\psi)+(\vec{\nabla}\psi^\dagger)\cdot\vec{\alpha}\psi\Big)=0\qquad\Rightarrow\quad\frac{\partial}{\partial t}(\psi^\dagger\psi)+\vec{\nabla}\cdot(\psi^\dagger\vec{\alpha}\psi)=0 \tag{3.33}$$

This equation is very similar to the Equation of continuity or the current conservation equation. Thus we can substitute, $\psi^\dagger\psi=\rho$ and $\psi^\dagger\vec{\alpha}\psi=\vec{J}$, which results in,

$$\frac{\partial\rho}{\partial t}+\vec{\nabla}\cdot\vec{J}=0 \tag{3.34}$$

Here the density, $\rho\geq 0$ is positive definite. And the current is given by, $\vec{J}=\psi^\dagger\vec{\alpha}\psi$. In covariant notation we can write, $J^\mu=\psi^\dagger\gamma^\mu\psi=(\rho,\vec{J})$, and hence the continuity equation becomes,

$$\partial_\mu J^\mu=0 \tag{3.35}$$

The quantity ρ can be thought of as either probability density, or number density, or any type of the charge density (weak, electric, weak iso-spin, etc), and $\vec{J}$ is the corresponding current of ρ. Thus Dirac equation predicts different types of charge densities and corresponding currents are conserved.

3.2.2 Linear Momentum and its equation of motion

By Ehrenfest's theorem, we know that the time evolution of the expectation value of a physical observable is related to the Hamiltonian operator by the relation,

$$\frac{d\langle\hat{\mathbf{\Omega}}\rangle}{dt}=\langle i[\hat{\mathbf{H}},\hat{\mathbf{\Omega}}]\rangle \tag{3.36}$$

Time rate of position operator

To get the time rate of the position operator, substitute $\hat{\mathbf{r}}$ in place of $\hat{\mathbf{\Omega}}$. Then we get,

$$\frac{d\langle\hat{\mathbf{r}}\rangle}{dt}=\langle i[\hat{\mathbf{H}},\hat{\mathbf{r}}]\rangle=\langle i[(\vec{\alpha}\cdot(i\vec{\nabla})+\underbrace{\cancel{\beta m}}_{\text{constant}}),\hat{\mathbf{r}}]\rangle=\langle\vec{\alpha}\cdot\vec{\nabla}(r)-r(\vec{\alpha}\cdot\vec{\nabla})\rangle=\langle\vec{\alpha}\cdot\big(\cancel{r\vec{\nabla}}+1\big)-\cancel{r\vec{\alpha}\cdot\vec{\nabla}}\rangle=\langle\vec{\alpha}\rangle$$

$$\Rightarrow\qquad\frac{d\langle\hat{\mathbf{r}}\rangle}{dt}=\langle i[\hat{\mathbf{H}},\hat{\mathbf{r}}]\rangle=\langle\vec{\alpha}\rangle\qquad\Rightarrow\qquad\text{Velocity operator }=\vec{\alpha} \tag{3.37}$$

Thus the rate of change of position operator is equal to the α vector which can be thought of as the velocity operator. And according to Eq.(3.28b), $\vec{\alpha}$ has Eigen values of ± 1 and hence in SI units they are $\pm c$.

Time rate of Linear momentum operator

For this, we shall substitute $i\vec{\nabla}$ in place of $\hat{\mathbf{\Omega}}$ in Ehrenfest's theorem. Then we get,

$$\frac{d\langle\hat{\mathbf{P}}\rangle}{dt} = \frac{d\langle i\vec{\nabla}\rangle}{dt} = \langle i[\hat{\mathbf{H}}, i\vec{\nabla}]\rangle = \langle\hat{\mathbf{H}}\vec{\nabla} - \vec{\nabla}\hat{\mathbf{H}}\rangle = \langle(\vec{\alpha}\cdot\vec{\nabla})\vec{\nabla} - \vec{\nabla}(\vec{\alpha}\cdot\vec{\nabla})\rangle \qquad \Rightarrow \qquad \frac{d\langle\hat{\mathbf{P}}\rangle}{dt} = 0 \tag{3.38}$$

Note that we can use just $\vec{\alpha}\cdot\vec{P}$ or $\vec{\alpha}\cdot\vec{\nabla}$ in place of Hamiltonian in commutator brackets as the term βm is a constant that will cancel. Now, if any operator $\hat{\mathbf{\Omega}}$ commutes with the Hamiltonian operator, then that physical quantity corresponding to the operator $\hat{\mathbf{\Omega}}$ is conserved. Since the linear momentum commutes with the Hamiltonian operator, we can say that Linear momentum is a conserved quantity. And that is expected since the law of conservation of momentum has to hold even at relativistic speeds as it is a universal law.

Anti-Commutator of Hamiltonian and velocity operator

$$\left\{\hat{\mathbf{H}}, \frac{d\vec{r}}{dt}\right\} = \{\hat{\mathbf{H}}, \vec{\alpha}\} = \{(\vec{\alpha}\cdot\vec{P}), \vec{\alpha}\} = (\vec{\alpha}\cdot\vec{P})\vec{\alpha} + \vec{\alpha}(\vec{\alpha}\cdot\vec{P}) = \vec{\alpha}^2\vec{P} + (\vec{\alpha})^2\vec{P} = \vec{P} + \vec{P} = 2\vec{P} \tag{3.39}$$

$$\Rightarrow \qquad \{(\vec{\alpha}\cdot\vec{P}), \frac{d\vec{r}}{dt}\} = 2\vec{P} \qquad \Rightarrow \qquad \langle\vec{\alpha}\rangle = \frac{\langle\vec{P}\rangle}{E} \tag{3.40}$$

This relation signifies the peculiarity of the velocity operator. Because we know that the individual components of position and linear momentum commute with each other, i.e., $[x_i, x_j] = [P_i, P_j] = 0 \quad (i, j = 1, 2, 3)$ and also we have $[x_i, \hat{\mathbf{P}}_j] = 0$ whenever $i \neq j$. But for the velocity operator, we have a constraint because of Clifford's algebra. We know that when $i \neq j$, by Eq.(3.28a) we get,

$$\{\alpha_i, \alpha_j\} = 0 \qquad \Rightarrow \qquad \alpha_i\alpha_j + \alpha_j\alpha_i = 0 \qquad \Rightarrow \qquad \alpha_j\alpha_i = -\alpha_i\alpha_j \tag{3.41}$$

$$\Rightarrow \qquad [\alpha_i, \alpha_j] = \alpha_i\alpha_j - \alpha_j\alpha_i = \alpha_i\alpha_j - (-\alpha_i\alpha_j) = 2\alpha_i\alpha_j \qquad \Rightarrow \qquad [\alpha_i, \alpha_j] \neq 0 \tag{3.42}$$

Thus different components of the velocity operator don't commute with each other. So, the wave function cannot be localized at all no matter what. Thus wave function appears to be smeared in space.

The Eigen values of the velocity operator are $\pm c$, this implies that ψ moves with the speed of light. But particle as a whole is not going at speed c. There is a phenomenon called '*zitterbewegung*' that happens in the wave function. This means that the wave function has a '*Jittery motion*' which oscillates back and forth as the particle moves. Though the wave function has an internal oscillation of velocity c, due to the jittery motion, the particle's instantaneous velocity will be lesser than c, which is its observed velocity.

3.2.3 Angular Momentum and emergence of Spin

Angular momentum is given by $\vec{L} = \vec{r}\times\vec{P}$. According to general Ehrenfest's theorem, its time rate is,

$$\frac{d\langle\vec{L}\rangle}{dt} = \langle i[\hat{\mathbf{H}}, \vec{L}]\rangle = \langle i[(\vec{\alpha}\cdot\vec{P}), \vec{r}\times\vec{P}]\rangle = \langle\vec{\alpha}\times\vec{P}\rangle \neq 0 \tag{3.43}$$

The commutator when worked turns out to be $\vec{\alpha}\times\vec{P}$. This anomaly is due to the fact that in relativistic quantum mechanics the velocity operator is a bit peculiar in terms of commutation relations as seen before. Thus we see that the angular momentum given by $\vec{L} = \vec{r}\times\vec{P}$ is not a conserved quantity in general. So we need to somehow modify the angular momentum expression and make it a conserved quantity. Thus we shall define total angular momentum as $\vec{J} = \vec{L} + \vec{S}$, where $\vec{S}$ is a quantity which has to be determined such that the time rate of $\langle\vec{J}\rangle$ has to be zero. So the time evolution of the expectation value of this quantity described by $\vec{S}$ has to compensate the cross product term between $\vec{\alpha}$ and $\vec{P}$. In Dirac basis,

$$\vec{S} = \frac{1}{2}\begin{pmatrix}\vec{\sigma} & 0 \\ 0 & \vec{\sigma}\end{pmatrix} \qquad (\vec{\sigma} \text{ are the Pauli matrices}) \tag{3.44}$$

$$\Rightarrow \qquad \frac{d\langle\vec{S}\rangle}{dt} = \left\langle i\left[\begin{pmatrix}0 & \vec{\sigma} \\ \vec{\sigma} & 0\end{pmatrix}\cdot\vec{P}, \frac{1}{2}\begin{pmatrix}\vec{\sigma} & 0 \\ 0 & \vec{\sigma}\end{pmatrix}\right]\right\rangle = \left\langle i\left[\begin{pmatrix}0 & \vec{\sigma}\cdot\vec{P} \\ \vec{\sigma}\cdot\vec{P} & 0\end{pmatrix}, \frac{1}{2}\begin{pmatrix}\vec{\sigma} & 0 \\ 0 & \vec{\sigma}\end{pmatrix}\right]\right\rangle \tag{3.45}$$

The commutator can be solved using an important property of Pauli matrices which is $\sigma_i\sigma_j = \delta_{ij} + i\varepsilon_{ijk}\sigma_k$ where ε_{ijk} is the Levi-Civita symbol which is equal to 1 for even permutations of ijk and -1 for odd permutations of ijk and 0 otherwise. Solving the commutator we get the relation,

$$\frac{d\langle\vec{S}\rangle}{dt} = -\langle\vec{\alpha}\times\vec{P}\rangle \tag{3.46}$$

$$\vec{J} = \vec{L} + \vec{S} \qquad \Rightarrow \quad \frac{d\langle\vec{J}\rangle}{dt} = \frac{d\langle\vec{L}\rangle}{dt} + \frac{d\langle\vec{S}\rangle}{dt} = \langle\vec{\alpha}\times\vec{P}\rangle - \langle\vec{\alpha}\times\vec{P}\rangle = 0 \qquad \Rightarrow \quad \frac{d\langle\vec{J}\rangle}{dt} = 0 \tag{3.47}$$

Thus the Total Angular Momentum ($\vec{J}$) is now conserved, which is got by modifying Angular momentum ($\vec{L}$) by adding another angular momentum quantity to it called $\vec{S}$. Now we see that even when the particle is at rest or if the particle has a constant linear momentum, i.e., if $\vec{L} = \vec{r}\times\vec{P} = 0$, the particle has some non zero angular momentum intrinsic to it called the spin angular momentum $\vec{S}$. It doesn't mean that the particle is physically spinning about an axis rather this is purely a relativistic quantum mechanical phenomenon that requires to be considered along with $\vec{L}$ so that the total angular momentum $\vec{J}$ is conserved. In Section(1.2.6), we saw the NRQM basis of spin. There the need for spin was not known. But through RQM, the intrinsic angular momentum emerged because of the Dirac equation. The eigen values of $\vec{S}$ is always $\pm 1/2$. Thus the Dirac theory explains Spin-half particles (fermions) only. The Spin Operators are,

$$\hat{\mathbf{\Sigma}}_x = \frac{1}{2}\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix} \qquad \hat{\mathbf{\Sigma}}_y = \frac{1}{2}\begin{pmatrix}0&-i&0&0\\i&0&0&0\\0&0&0&-i\\0&0&i&0\end{pmatrix} \qquad \hat{\mathbf{\Sigma}}_z = \frac{1}{2}\begin{pmatrix}1&0&0&0\\0&-1&0&0\\0&0&1&0\\0&0&0&-1\end{pmatrix} \tag{3.48}$$

To get the $z-$component of spin angular momentum, we need to operate $\hat{\mathbf{\Sigma}}_z$ on the spinor ψ which gives,

$$\hat{\mathbf{\Sigma}}_z\psi = \frac{1}{2}\begin{pmatrix}1&0&0&0\\0&-1&0&0\\0&0&1&0\\0&0&0&-1\end{pmatrix}\begin{pmatrix}\psi_1\\\psi_2\\\psi_3\\\psi_4\end{pmatrix} = \frac{1}{2}\begin{pmatrix}\psi_1\\-\psi_2\\\psi_3\\-\psi_4\end{pmatrix} \begin{matrix}\rightarrow & \text{Spin up}\\ \rightarrow & \text{Spin down}\\ \rightarrow & \text{Spin up}\\ \rightarrow & \text{Spin down}\end{matrix} \tag{3.49}$$

Here the positive sign corresponds to the spin-up state. And negative sign correspond to spin-down state. Thus the components ψ_1 and ψ_3 represent the spin up particles and ψ_2 and ψ_4 represent the spin down particles. Since ψ has two spin-up and spin-down states, it is called the Bi-Spinor.

3.2.4 Helicity

The helicity of a particle is the component of the Total angular momentum $\vec{J} = \vec{L} + \vec{S}$ in the direction of the propagation of the particle. We know that The direction of propagation of the particle will be along with momentum $\vec{P}$ always. If $|\vec{P}| = P$ and $\hat{P}$ is the unit vector along the direction of momentum, we get,

$$\text{Helicity Operator} = \hat{\mathbf{h}} = \vec{J}\cdot\frac{\vec{P}}{|P|} = \left(\vec{L}+\vec{S}\right)\cdot\hat{P} = \underbrace{\vec{L}\cdot\hat{P}}_{0} + \vec{S}\cdot\hat{P} = \vec{S}\cdot\hat{P} \tag{3.50}$$

The Commutator of Hamiltonian $\hat{\mathbf{H}}$ and helicity $\hat{\mathbf{h}}$ is zero since the matrices of these operators commute.

$$[\hat{\mathbf{H}},\hat{\mathbf{h}}] = \frac{1}{|\vec{P}|}[\vec{\alpha}\cdot\vec{P},\vec{S}\cdot\vec{P}] = \frac{1}{2|\vec{P}|}\left[\begin{pmatrix}0 & \vec{\sigma}\cdot\vec{P}\\ \vec{\sigma}\cdot\vec{P} & 0\end{pmatrix}, \begin{pmatrix}\vec{\sigma}\cdot\vec{P} & 0\\ 0 & \vec{\sigma}\cdot\vec{P}\end{pmatrix}\right] = 0 \tag{3.51}$$

Thus the helicity of a particle is a conserved quantity and doesn't change with time. Yet, it is not a Lorentz invariant because the vectors change when measured at high speeds if a Lorentz boost is used.

$$\text{Consider} \quad h^2 = \hat{\mathbf{h}}\hat{\mathbf{h}} = (\vec{S}\cdot\hat{P})(\vec{S}\cdot\hat{P}) = \frac{1}{4}\begin{pmatrix}\vec{\sigma}\cdot\hat{P} & 0\\ 0 & \vec{\sigma}\cdot\hat{P}\end{pmatrix}\begin{pmatrix}\vec{\sigma}\cdot\hat{P} & 0\\ 0 & \vec{\sigma}\cdot\hat{P}\end{pmatrix} = \frac{1}{4}\begin{pmatrix}|\hat{P}|^2 & 0\\ 0 & |\hat{P}|^2\end{pmatrix} = \frac{I}{4} \tag{3.52}$$

$$\Rightarrow \qquad \text{Eigen Values of Helicity,} \quad h = \pm\frac{1}{2} \tag{3.53}$$

Thus we see that for any particle the helicity is either positive or negative half. The particles with positive helicity are called as right-handed particles (spin and momentum are in similar direction). The particles with negative helicity are called left-handed particles whose spin and momentum are in dissimilar directions.

3.3 Plane wave Solutions for a free particle

These are the solutions of the Dirac equation whose form is a general plane wave described by a wave wiggle factor (e^{-iEt}). By this spinor notation, the Dirac equation becomes a set of four simultaneous differential equations where the dependent variables are the components of spinor ψ. Using $\partial_\mu = (\partial_t, \partial_x, \partial_y, \partial_z)$ we get,

$$i\partial_t\psi_1 + i\partial_x\psi_4 + \partial_y\psi_4 + i\partial_z\psi_3 - m\psi_1 = 0 \tag{3.54a}$$

$$i\partial_t\psi_2 + i\partial_x\psi_3 - \partial_y\psi_3 - i\partial_z\psi_4 - m\psi_2 = 0 \tag{3.54b}$$

$$i\partial_t\psi_3 + i\partial_x\psi_2 + \partial_y\psi_2 + i\partial_z\psi_1 + m\psi_3 = 0 \tag{3.54c}$$

$$i\partial_t\psi_4 + i\partial_x\psi_1 - \partial_y\psi_1 - i\partial_z\psi_2 + m\psi_4 = 0 \tag{3.54d}$$

In NRQM, the normalization condition was $\psi^\dagger\psi = 1$. But here because of Lorentz contraction, the three-volume element $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ will be contracted and hence the normalization condition becomes,

$$\psi^\dagger\psi = \frac{|E|}{m} \qquad \text{and} \qquad \overline{\psi}\psi = 1 \tag{3.55}$$

Case 1: Free particle at rest $\rightarrow P_x = P_y = P_z = 0$

Now since the particle is at rest we get all the derivatives of the spinor components with respect to x, y, z to be zero. So the set of partial differential equations becomes,

$$i\partial_t\psi_1 = m\psi_1 \quad , \quad i\partial_t\psi_2 = m\psi_2 \quad , \quad i\partial_t\psi_3 = m\psi_3 \quad , \quad i\partial_t\psi_4 = m\psi_4 \tag{3.56}$$

As the ψ_i's are independent we can solve the equations exclusively, i.e., if $\psi_i \neq 0 \quad \Rightarrow \quad \psi_j = 0 \ (\forall \ i \neq j)$.

$$\underbrace{\overbrace{\psi = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} e^{-imt}}^{\text{Spin-up state}} \quad \text{or} \quad \overbrace{\psi = \begin{pmatrix}0\\1\\0\\0\end{pmatrix} e^{-imt}}^{\text{Spin-down state}}}_{E=m \ \rightarrow \ \text{Positive energy}} \quad \text{or} \quad \underbrace{\overbrace{\psi = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} e^{imt}}^{\text{Spin-up state}} \quad \text{or} \quad \overbrace{\psi = \begin{pmatrix}0\\0\\0\\1\end{pmatrix} e^{imt}}^{\text{Spin-down state}}}_{E=-m \ \rightarrow \ \text{Negative energy}} \tag{3.57}$$

Thus we get to see four different solutions where two solutions correspond to positive energy ($E = m$) which is for the spin up and down state. And other two solutions correspond to the negative energy ($E = -m$) with spin up and down state. Since the spinor is separable into different forms which correspond to one particular spin state we can tell that a particle at rest can be in pure up-spin or pure down-spin state.

Case 2: Free particle moving in $x-$axis $\rightarrow P_y = P_z = 0$

Since the particle is moving in the $x-$axis only, we get the derivatives of the component with respect to y and z to be zero. So, we shall set the terms having ∂_y and ∂_x to zero. Here the components ψ_i are not independent as in previous case. ψ_1 and ψ_4 are not separable, i.e., if ψ_1 is considered then there will be a ψ_4 component. And similarly, ψ_2 and ψ_3 are not separable. And the positive energy solutions are,

$$\psi = \begin{pmatrix}1\\0\\0\\\dfrac{P_x}{E+m}\end{pmatrix} e^{-i(P_x - Et)} \qquad \text{or} \qquad \psi = \begin{pmatrix}0\\1\\\dfrac{P_x}{E+m}\\0\end{pmatrix} e^{-i(P_x - Et)} \tag{3.58}$$

The total energy of the free particle is $E = \sqrt{P_x^2 - m^2}$. Then the negative energy solutions are,

$$\psi = \begin{pmatrix}0\\\dfrac{-P_x}{|E|+m}\\1\\0\end{pmatrix} e^{-i(P_x - Et)} \qquad \text{or} \qquad \psi = \begin{pmatrix}\dfrac{-P_x}{|E|+m}\\0\\0\\1\end{pmatrix} e^{-i(P_x - Et)} \tag{3.59}$$

The total energy of the particles is negative and is given by $E = -\sqrt{P_x^2 - m^2}$. This we see that the energy can have two values and written as $E^2 = P_x^2 + m^2$, which is obvious as the total energy of a free particle in motion is described the same formula according to special relativity. Here the spinors are having both spin-up and spin-down states present in the solutions corresponding to both positive energy particle and negative energy type of particle. So for a particle in motion, it is not possible to be in a pure spin-up or spin-down state since the spinor describing it has both spin-up component and spin-down component. The significance of particles with Positive and Negative energy will be dealt in detail in Chapter(4)

3.4 The Dirac Lagrangian

The Dirac Lagrangian is constructed such that, if the Euler-Lagrange equation of motion of the system is derived, then it has to yield the Dirac Equation and its hermitian adjoint equation. So the appropriate Dirac Lagrangian is given by,

$$\boxed{\mathcal{L} = \overline{\psi}(i\partial\!\!\!/ - m)\psi} \tag{3.60}$$

When the Euler Lagrange equation of motion is used separately for ψ and $\overline{\psi}$, i.e.,

$$\frac{\partial \mathcal{L}}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \right) = 0 \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial \overline{\psi}} - \partial^\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial^\mu \overline{\psi})} \right) = 0 \tag{3.61}$$

When the equation of motion is solved with respect to $\overline{\psi}$ we get the Dirac equation, i.e., $(i\partial\!\!\!/ - m)\psi = 0$. And when the equation is solved with respect to ψ, then we get the hermitian adjoint of the Dirac equation, i.e., $\overline{\psi}(i\overleftarrow{\partial\!\!\!/} + m) = 0$. Thus the Lagrangian we chose is the correct one and very useful Lagrangian because it contains much information about the spin $n/2$ particles (fermions) and the quantities they conserve.

3.4.1 Symmetries in Dirac Lagrangian

space-time translations

The Lagrangian is invariant under linear space-time translations where $\delta\psi \to \varepsilon^\mu \partial_\mu \psi$. This gives rise to,

$$\text{Energy tensor} \;=\; T^{\mu\nu} = i\overline{\psi}\gamma^\mu \partial^\nu \psi \tag{3.62}$$

$$\text{Conserved quantity} = \text{Energy} \;=\int T^{00}\, \mathrm{d}^3x = \int i\overline{\psi}\gamma^0 \frac{\partial \psi}{\partial t}\, \mathrm{d}^3x = \int i\psi^\dagger \gamma^0 \big(-i\gamma^j \partial_j + m \big)\psi \, \mathrm{d}^3x \tag{3.63}$$

Thus the total energy of the system is a conserved quantity which arises due to the fact that the Lagrangian is invariant when space-time translation is done on the ψ function.

Lorentz Transformation

Lorentz transformation of the Dirac Lagrangian is, $$\delta\psi^\alpha = \omega^{\mu\nu}\left[\frac{1}{2}\big(S^{\mu\nu}\big)^\alpha_{\ \beta}\psi^\beta - x_\nu \partial_\mu \psi^\alpha\right] \tag{3.64}$$

The conserved quantity will be the total Angular momentum four-vector $\mathcal{J}^\mu$ which has an extra time component. The spatial components are the usual $\vec{L} = \vec{r} \times \vec{P}$ and the extra term corresponds to the time component of the four angular momenta which provides the particles with an intrinsic angular momentum giving rise to 'Spin'. The quantization of the Dirac spinor gives rise to a spin $1/2$ particle.

Phase Rotation Symmetry

When the spinor is subject to rotation under the phase transformation $\delta\psi = e^{-i\varepsilon}\psi$, the Lagrangian remains invariant and gives rise to a four current vector j^μ which is conserved according to the equation of continuity.

$$j^\mu = \psi^\dagger \gamma^\mu \psi \quad \Rightarrow \quad \partial_\mu j^\mu = 0 \tag{3.65}$$

Chapter 4

Matter and Anti Matter

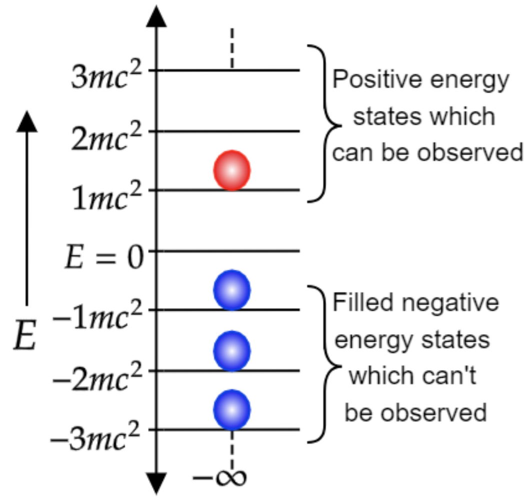
We see that whenever we try to unify quantum mechanics with relativistic mechanics, the solutions always yield two types of energy states one is a particle with positive rest energy and the other is a particle with negative rest energy. Klein Gordon's equation first predicted the negative energy solutions. Dirac tried to eliminate the negative energy solution by trying to linearize the mass-Equivalence relation. But even there the result predicted energy of a particle can be both positive and negative. The positive rest energy particles are normal and are found everywhere in the universe. Electron, proton, neutron, etc are all positive rest energy normal particles. But the negative energy particles are called the Anti-matter particles which are counter particles for normal matter particles. For example, the anti-particle of an electron is a positron. The electric charge density of electrons is negative conventionally. But for positron it is positive. Thus the electric charge is opposite with the same mass for particle and anti-particle pairs. Not only electric charge but every known charge like the weak charge, color charge, flavor charges, etc are opposite in a particle and anti-particle pair, except that their mass will be the same.

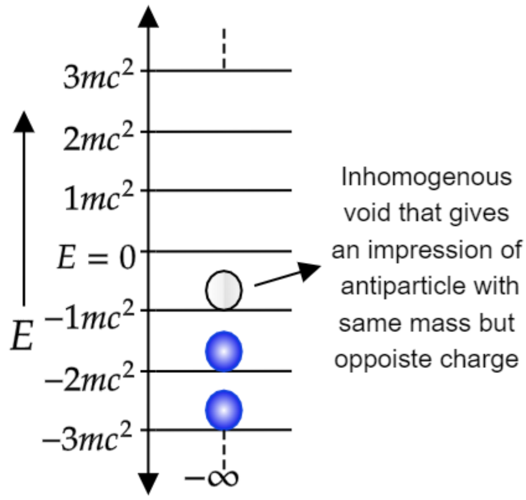
If we recall the spinors used in section(1.2.6) there we had just two components for a particle that described up spin and down spin state. But through the Dirac equation, we got a spinor that has 4 components. Where the first two components of ψ, describe the spin up and down state of a particle with mass m and the next two components represent the spin up and down state of the anti-particle which has same the mass m. Thus the spinor describes both the particle and its anti-particle and that information are encoded in the ψ. Dirac explained this concept by an electron sea theory.

4.1 Dirac Sea Theory of Electrons

Since the negative rest energy states are allowed this implies that there can be infinite negative energy states of $E = -mc^2, -2mc^2, -3mc^2, \ldots, \infty$. All these infinite negative energy states have been occupied by an infinite number of electrons and form a sea of negative energy electrons that is present everywhere in the universe. They can NOT be observed because they occupy the negative energy states homogeneously with only one electron per negative energy state because of Pauli's Exclusion principle, this makes the Dirac sea completely homogeneous and impossible to detect. But we can observe the electrons that have positive rest energy because the positive rest energy states are not filled completely and is inhomogeneous.

If there is a void in the electron sea containing negative rest energy electrons, then the void forms an inhomogeneity in the electron sea which can be observed. The void is called a hole that can move in the Dirac sea. It has the inertia and mass as that of an electron. But the hole acts as if it has a positive charge since a negative electric charge has been missing in the electron sea. This hole can be thought of as the anti-particle of the electron called as 'Positron' (e^+) which resides in the electron sea as a void and can be observed. Electron and positron can have two different states of spin each. This idea emerged elegantly in the Dirac equation as a spinor with four components, two for e^- and the other two for e^+. So electrons and positrons cannot exist without each other and are like the two sides of the same coin. They are positive and negative energy solutions of the same type of particles in a Dirac sea.

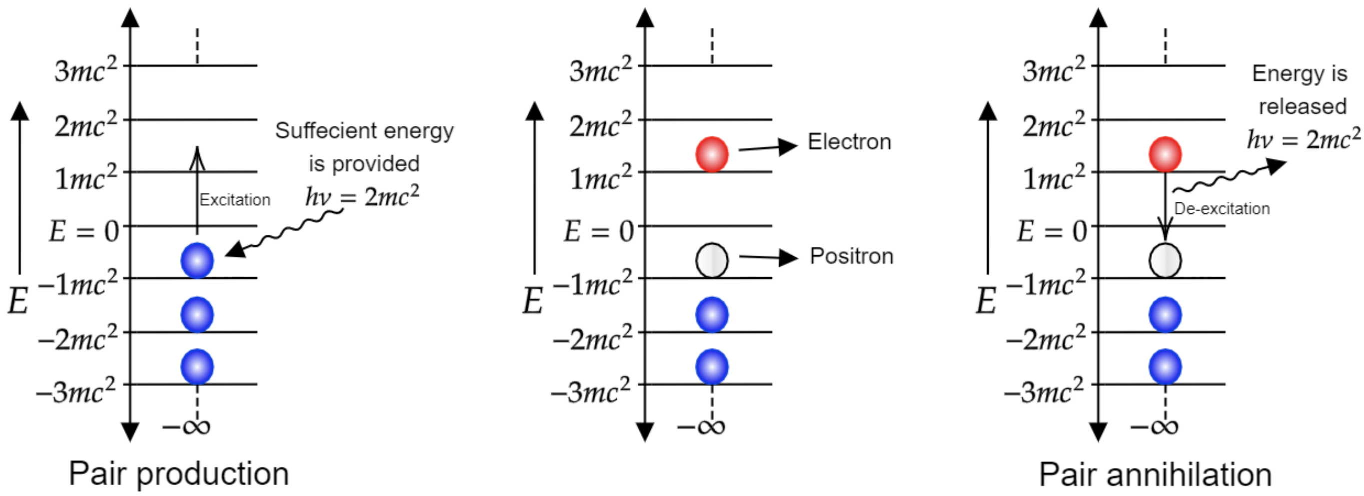




4.1.1 Pair production and pair Annihilation

The phenomenon of pair production and pair annihilation was elegantly explained by Dirac through the electron sea theory. Consider a region of space with no electrons with positive energy in it. There will be electrons with negative rest energy which cant be observed. So the region will be like a vacuum with no particle present in it. But if we want to observe an electron that is at $-1mc^2$ energy then we need to provide an energy of $2mc^2$ because only then it can be transited from the $-1mc^2$ negative rest energy level to the $1mc^2$ positive rest energy level. Then the result of this process is that an electron in $+1mc^2$ energy and a positron (void in electron sea) in $-1mc^2$ energy state are produced. This process has just created a particle-antiparticle pair which is called the pair production.

Now, consider that there is an electron-positron pair in a region of space. If the electron comes in contact with the positron (void), it can transit back to the $-1mc^2$ energy level by releasing energy equal to $2mc^2$, and then that electron cannot be observed now as it is in negative energy level. Thus the region of space has no particles in it and has become a vacuum. Hence we can say that here two particles have been annihilated and released an equivalent amount of energy. This is the phenomenon of pair annihilation.



Though the Dirac sea theory explains most of the aspects of an electron, it is a theoretical artifact that is not real because there is no existence of the Dirac sea. The idea of the sea is modified into the field whose quantum is the particle which the field is describing. This is the main idea of the so-called '*Quantum field theory*'. So here the Dirac sea of particles is replaced by the quantum fields of elementary particles where the particle-antiparticle counterparts are explained as the $+ve$ and $-ve$ energy vibration. But still, it was Dirac who first predicted Anti-matter and gave a basic insight to the sea (field) of elementary particles as the idea of electron sea can be extended to other elementary particles having their own sea.

4.2 Properties of particles Vs anti particles

4.2.1 Opposite Charges

In relativistic quantum particle physics, there are many associated charges for a particle concerning the type of interactions they have. Some important types of charges are, **Electric charge** (Electro-magnetic force and interaction with photons), **Colour Charge** (Strong nuclear force and interaction with gluons), **Weak Charge** (Weak nuclear forces and interaction with W and Z bosons), **Weak Hypercharge** (Relating electric charge and iso-spin of the particle), **Flavour Charges** (Each type of particle has an associated quantum number of its own like Baryon number, Strange number, Charm number, Lepton number, Muon number, etc). So for particle and anti-particle all these charges will have opposite signs. But if there is a particle with all these charges and quantum numbers to be zero, then the particle will be its own anti-particle and are called 'Truly neutral particles'. (Example:- Photon, Z-Boson, Higgs Boson).

4.2.2 Opposite Vibrational Energies

As said earlier, according to Quantum field theory each elementary particle has its own quantum field. And these fields normally have zero energy. And the regions of space where the field has excited vibrational energy state, that vibrational energy corresponds to the quantum particle and its anti-particle of that quantum field. If we choose the normal every day particles we encounter to have positive energy states, then their anti-matter counterparts will have opposite energy states but same frequency of vibration. For example, Electrons and positrons have a field associated with them, Up-quark and anti Up-quark has its own field, Muon neutrino and anti muon neutrino have their own fields, etc. Moreover, the interaction between an elementary and their elementary particle is fundamentally because of the interaction between their quantum fields according, 'Interacting quantum field theory'.

4.2.3 Helicity

As seen in Section(3.2.4), helicity is the projection of the total angular momentum of the particle in the direction of motion (linear momentum). Also the helicity operator has two orthogonal eigen states and the eigen values are $\pm 1/2$. So for any particle its helicity will be either $+1/2$ or $-1/2$. So, if a particle has helicity to be $+1/2$, then it is called as Right handed particles. And if a particle has $-1/2$ helicity, then the particle is left handed. Now, all matter particles we have observed are left handed. And all anti-matter particles we have observed yet are right handed. Note: Chirality of particles on other hand is different and that deals with the transformation (left handed or right handed) of the particle in the Poincaré group.

Conclusion

In this term paper, I have studied the basic ideas of Relativistic Quantum Mechanics. We get to see many results and predict new phenomenon when we try to unify the theory of special relativity with quantum mechanics. Klein Gordon equation predicts the behavior of massless particles like a photon or massive particles with zero-spin like Higgs Boson, π meson, etc. Dirac theory provides a very satisfactory explanation for spin $1/2$ particles like electrons. Dirac explained so many phenomena of atoms that could not be explained by Schrodinger's theory like the Fine structure, Zeeman effect, Stark effect, Spin related phenomenon, etc. One of the major predictions of Dirac theory includes the existence of Antimatter which was described by Dirac in the electron sea model, the idea which lead to the formulation of Quantum Field Theory. Relativistic Quantum Mechanics is a very broad topic and is one of the initial concepts which lead to the more advanced concepts like Quantum Field Theory and Particle Physics. Further, the unification of these concepts with the General Theory of Relativity will lead to the String theory/Loop Quantum Gravity.

Further studies after this term paper regarding advanced RQM may include formulating electromagnetic field interaction with Dirac electron using local gauge-invariant theory, Foldy-Wouthuysen transformation, Zitterbewegung, Weyl and Majorana Theory, Lorentz and Poincaré groups, CPT symmetry, etc.

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